

asset-class-return-correlations

May 2, 2026

0.1 Introduction

In this post, we'll take a look at historical return correlations between asset classes, including rolling returns, and how those correlations have changed over time.

Python Imports

```
[1]: # Standard Library
import datetime
import io
import os
import random
import sys
import warnings

from datetime import datetime, timedelta
from pathlib import Path

# Data Handling
import numpy as np
import pandas as pd

# Data Visualization
import matplotlib.dates as mdates
import matplotlib.pyplot as plt
import matplotlib.ticker as mtick
import seaborn as sns
from matplotlib.ticker import FormatStrFormatter, FuncFormatter, MultipleLocator

# Data Sources
import yfinance as yf
import pandas_datareader.data as web

# Statistical Analysis
import statsmodels.api as sm

# Machine Learning
from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler
```

```
# Suppress warnings
warnings.filterwarnings("ignore")
```

```
[2]: # Add the source subdirectory to the system path to allow import config from
      ↪ settings.py
current_directory = Path(os.getcwd())
BASE_DIR = current_directory.parent.parent.parent
src_directory = BASE_DIR / "src"
sys.path.append(str(src_directory)) if str(src_directory) not in sys.path else
      ↪ None

# Import settings.py
from settings import config

# Add other configured directories
CONTENT_DIR = config("CONTENT_DIR")
POSTS_DIR = config("POSTS_DIR")
PAGES_DIR = config("PAGES_DIR")
PUBLIC_DIR = config("PUBLIC_DIR")
SOURCE_DIR = config("SOURCE_DIR")
DATA_DIR = config("DATA_DIR")
DATA_MANUAL_DIR = config("DATA_MANUAL_DIR")
```

0.2 Python Functions

Here are the functions needed for this project:

- `load_data`: Load data from a CSV, Excel, or Pickle file into a pandas DataFrame.
- `pandas_set_decimal_places`: Set the number of decimal places displayed for floating-point numbers in pandas.
- `plot_histogram`: Plot the histogram of a data set from a DataFrame.
- `plot_scatter`: Plot the data from a DataFrame for a specified date range and columns.
- `plot_time_series`: Plot the timeseries data from a DataFrame for a specified date range and columns.
- `run_linear_regression`: Run a linear regression using statsmodels OLS and return the results.
- `summary_stats`: Generate summary statistics for a series of returns.
- `yf_pull_data`: Download daily price data from Yahoo Finance and export it.

```
[3]: from load_data import load_data
      from pandas_set_decimal_places import pandas_set_decimal_places
      from plot_heatmap import plot_heatmap
      from plot_histogram import plot_histogram
      from plot_scatter import plot_scatter
      from plot_time_series import plot_time_series
      from run_regression import run_regression
      from summary_stats import summary_stats
      from yf_pull_data import yf_pull_data
```

0.3 Data Overview

For this exercise, we will (mostly) use ETFs as a proxy for asset classes and will use the following:

- Large Cap US Stocks / S&P 500 – IVV (iShares S&P 500 ETF)
- Mid Cap US Stocks / S&P MidCap 400 – IJH (iShares S&P MidCap 400 ETF)
- Small Cap US Stocks / S&P SmallCap 600 – IJR (iShares S&P SmallCap 600 ETF)
- US Tech Stocks / Nasdaq 100 – QQQ (Invesco QQQ Trust, Series 1)
- US Value Stocks – IWD (iShares Russell 1000 Value ETF)
- US Growth Stocks – IWF (iShares Russell 1000 Growth ETF)
- International Developed Market Stocks – EFA (iShares MSCI EAFE ETF)
- Emerging Market Stocks – EEM (iShares MSCI Emerging Markets ETF)
- European Stocks – IEV (iShares S&P Europe 350 ETF)
- Short-Term US Treasuries – SHY (iShares 1-3 Year Treasury Bond ETF)
- Medium-Term US Treasuries – IEF (iShares 7-10 Year Treasury Bond ETF)
- Long-Term US Treasuries – TLT (iShares 20+ Year Treasury Bond ETF)
- Aggregate Bonds – AGG (iShares Core U.S. Aggregate Bond ETF)
- Gold – GLD (SPDR Gold Shares)
- Commodities – GSG (iShares S&P GSCI Commodity-Indexed Trust)
- Real Estate – IYR (iShares U.S. Real Estate ETF)
- Bitcoin – BTC-USD (Bitcoin USD)
- Ethereum – ETH-USD (Ethereum USD)

For all of these, we will use the adjusted closing price, which accounts for dividends and stock splits.

0.4 Acquire & Plot Data

We'll now pull the data for all the ETFs and cryptocurrencies listed above.

```
[4]: pandas_set_decimal_places(2)

# Create list of tickers to pull data for
us_equity_tickers = ["IVV", "IJH", "IJR", "QQQ", "IWB", "IWD", "IWF", "IWM"]
intl_equity_tickers = ["EFA", "EEM", "IEV"]
equity_tickers = us_equity_tickers + intl_equity_tickers
bond_tickers = ["SHY", "IEF", "TLT", "AGG"]
commodity_tickers = ["GLD", "GSG"]
real_estate_tickers = ["IYR"]
cryptoasset_tickers = ["BTC-USD", "ETH-USD"]
etf_tickers = us_equity_tickers + intl_equity_tickers + bond_tickers + \
    commodity_tickers + real_estate_tickers
tickers_dict = {
    "IVV" : "Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)",
    "IJH" : "Mid Cap US Stocks / S&P MidCap 400 -- IJH (iShares S&P MidCap 400_
    ETF)",
    "IJR" : "Small Cap US Stocks / S&P SmallCap 600 -- IJR (iShares S&P_
    SmallCap 600 ETF)",
    "QQQ" : "US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)",
    "IWB" : "Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)",
```

```

    "IWM" : "Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)",
    "IWD" : "Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value_
↪ETF)",
    "IWF" : "Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000_
↪Growth ETF)",
    "EFA" : "International Developed Market Stocks -- EFA (iShares MSCI EAFE_
↪ETF)",
    "EEM" : "Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)",
    "IEV" : "European Stocks -- IEV (iShares S&P Europe 350 ETF)",
    "SHY" : "Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond_
↪ETF)",
    "IEF" : "Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond_
↪ETF)",
    "TLT" : "Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond_
↪ETF)",
    "AGG" : "Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)",
    "GLD" : "Gold -- GLD (SPDR Gold Shares)",
    "GSG" : "Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)",
    "IYR" : "Real Estate -- IYR (iShares U.S. Real Estate ETF)",
    "BTC-USD" : "Bitcoin -- BTC-USD (Bitcoin USD)",
    "ETH-USD" : "Ethereum -- ETH-USD (Ethereum USD)",
}

```

```

[5]: for ticker in etf_tickers:
    yf_pull_data(
        base_directory=DATA_DIR,
        ticker=ticker,
        adjusted=False,
        source="Yahoo_Finance",
        asset_class="Exchange_Traded_Funds",
        excel_export=True,
        pickle_export=True,
        output_confirmation=False,
    )

for ticker in cryptoasset_tickers:
    yf_pull_data(
        base_directory=DATA_DIR,
        ticker=ticker,
        adjusted=False,
        source="Yahoo_Finance",
        asset_class="Cryptocurrencies",
        excel_export=True,
        pickle_export=True,
        output_confirmation=False,
    )

```

We'll then perform the following: * Load data * Rename columns to include the ticker (e.g. "QQQ_Close", "QQQ_Adj_Close", etc.) * Drop all columns except for "Adj Close" * Calculate the daily returns * Combine the data into a single DataFrame

```
[6]: fund_data = pd.DataFrame()

for ticker in etf_tickers:
    data = load_data(
        base_directory=DATA_DIR,
        ticker=ticker,
        source="Yahoo_Finance",
        asset_class="Exchange_Traded_Funds",
        timeframe="Daily",
        file_format="pickle",
    )

    # Rename columns to "QQQ_Close", etc.
    data = data.rename(columns={
        "Adj Close": f"{ticker}_Adj_Close",
        "Close": f"{ticker}_Close",
        "High": f"{ticker}_High",
        "Low": f"{ticker}_Low",
        "Open": f"{ticker}_Open",
        "Volume": f"{ticker}_Volume"
    })

    # Drop all columns except for the adjusted close price and date index
    data = data[[f"{ticker}_Adj_Close"]]

    # Calculate daily returns and add as new column
    data[f"{ticker}_Daily_Return"] = data[f"{ticker}_Adj_Close"].pct_change()

    # Concatenate the data for this ticker with the main fund_data DataFrame
    fund_data = pd.concat([fund_data, data], axis=1)

for ticker in cryptoasset_tickers:
    data = load_data(
        base_directory=DATA_DIR,
        ticker=ticker,
        source="Yahoo_Finance",
        asset_class="Cryptocurrencies",
        timeframe="Daily",
        file_format="pickle",
    )

    # Rename columns to "BTC-USD_Close", etc.
    data = data.rename(columns={
```

```

    "Adj Close": f"{ticker}_Adj_Close",
    "Close": f"{ticker}_Close",
    "High": f"{ticker}_High",
    "Low": f"{ticker}_Low",
    "Open": f"{ticker}_Open",
    "Volume": f"{ticker}_Volume"
})

# Drop all columns except for the adjusted close price and date index
data = data[[f"{ticker}_Adj_Close"]]

# Calculate daily returns and add as new column
data[f"{ticker}_Daily_Return"] = data[f"{ticker}_Adj_Close"].pct_change()

# Concatenate the data for this ticker with the main fund_data DataFrame
fund_data = pd.concat([fund_data, data], axis=1)

display(fund_data)

```

	IVV_Adj_Close	IVV_Daily_Return	IJH_Adj_Close	IJH_Daily_Return	\
Date					
1999-03-10	NaN	NaN	NaN	NaN	
1999-03-11	NaN	NaN	NaN	NaN	
1999-03-12	NaN	NaN	NaN	NaN	
1999-03-15	NaN	NaN	NaN	NaN	
1999-03-16	NaN	NaN	NaN	NaN	
...	...	...	...	...	
2026-04-27	718.50	0.00	72.84	0.00	
2026-04-28	714.96	-0.00	72.11	-0.01	
2026-04-29	714.89	-0.00	71.57	-0.01	
2026-04-30	722.07	0.01	72.77	0.02	
2026-05-01	NaN	NaN	NaN	NaN	

	IJR_Adj_Close	IJR_Daily_Return	QQQ_Adj_Close	QQQ_Daily_Return	\
Date					
1999-03-10	NaN	NaN	43.07	NaN	
1999-03-11	NaN	NaN	43.29	0.00	
1999-03-12	NaN	NaN	42.23	-0.02	
1999-03-15	NaN	NaN	43.44	0.03	
1999-03-16	NaN	NaN	43.81	0.01	
...	...	...	...	...	
2026-04-27	136.56	0.00	664.23	0.00	
2026-04-28	135.79	-0.01	657.55	-0.01	
2026-04-29	134.73	-0.01	661.57	0.01	
2026-04-30	137.10	0.02	667.74	0.01	
2026-05-01	NaN	NaN	NaN	NaN	

	IWB_Adj_Close	IWB_Daily_Return	...	GLD_Adj_Close	\
--	---------------	------------------	-----	---------------	---

Date			...	
1999-03-10	NaN		NaN	NaN
1999-03-11	NaN		NaN	NaN
1999-03-12	NaN		NaN	NaN
1999-03-15	NaN		NaN	NaN
1999-03-16	NaN		NaN	NaN
...	...	...	...	...
2026-04-27	391.14		0.00	429.89
2026-04-28	389.07		-0.01	421.91
2026-04-29	388.87		-0.00	417.41
2026-04-30	392.83		0.01	423.66
2026-05-01	NaN		NaN	NaN

	GLD_Daily_Return	GSG_Adj_Close	GSG_Daily_Return	IYR_Adj_Close	\
Date					
1999-03-10	NaN	NaN	NaN	NaN	NaN
1999-03-11	NaN	NaN	NaN	NaN	NaN
1999-03-12	NaN	NaN	NaN	NaN	NaN
1999-03-15	NaN	NaN	NaN	NaN	NaN
1999-03-16	NaN	NaN	NaN	NaN	NaN
...	...	...	...	...	...
2026-04-27	-0.01	32.88	0.01	100.83	
2026-04-28	-0.02	33.34	0.01	101.74	
2026-04-29	-0.01	34.56	0.04	100.90	
2026-04-30	0.01	34.39	-0.00	102.63	
2026-05-01	NaN	NaN	NaN	NaN	NaN

	IYR_Daily_Return	BTC-USD_Adj_Close	BTC-USD_Daily_Return	\
Date				
1999-03-10	NaN	NaN	NaN	
1999-03-11	NaN	NaN	NaN	
1999-03-12	NaN	NaN	NaN	
1999-03-15	NaN	NaN	NaN	
1999-03-16	NaN	NaN	NaN	
...	...	...	...	...
2026-04-27	-0.01	77366.62	-0.02	
2026-04-28	0.01	76350.67	-0.01	
2026-04-29	-0.01	75776.13	-0.01	
2026-04-30	0.02	76304.32	0.01	
2026-05-01	NaN	78179.00	0.02	

	ETH-USD_Adj_Close	ETH-USD_Daily_Return
Date		
1999-03-10	NaN	NaN
1999-03-11	NaN	NaN
1999-03-12	NaN	NaN
1999-03-15	NaN	NaN
1999-03-16	NaN	NaN

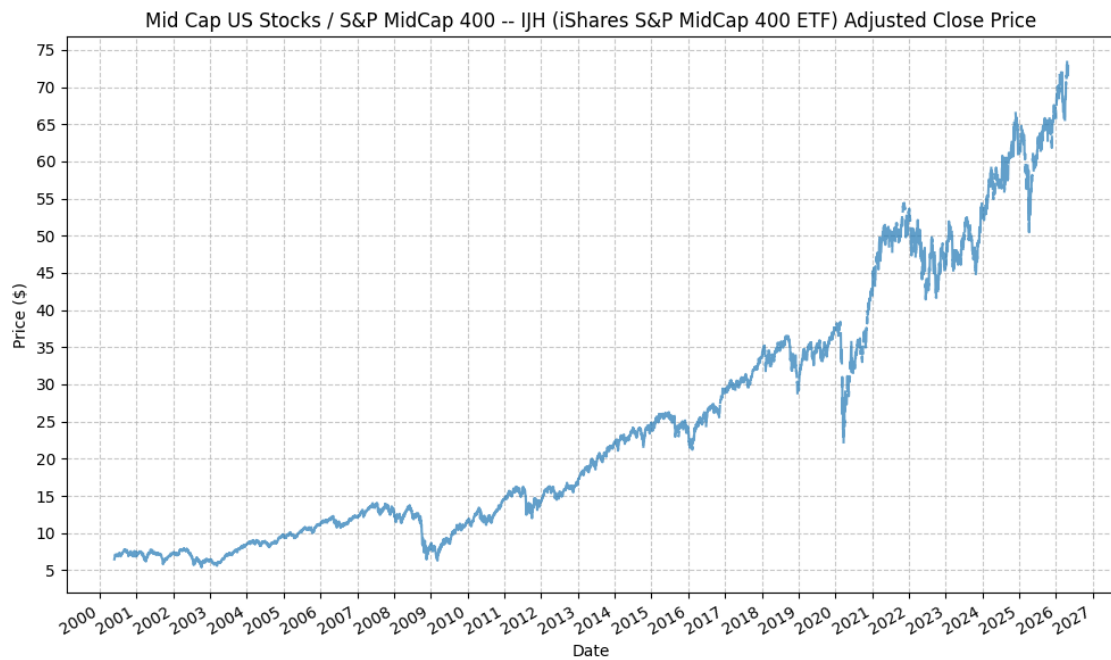
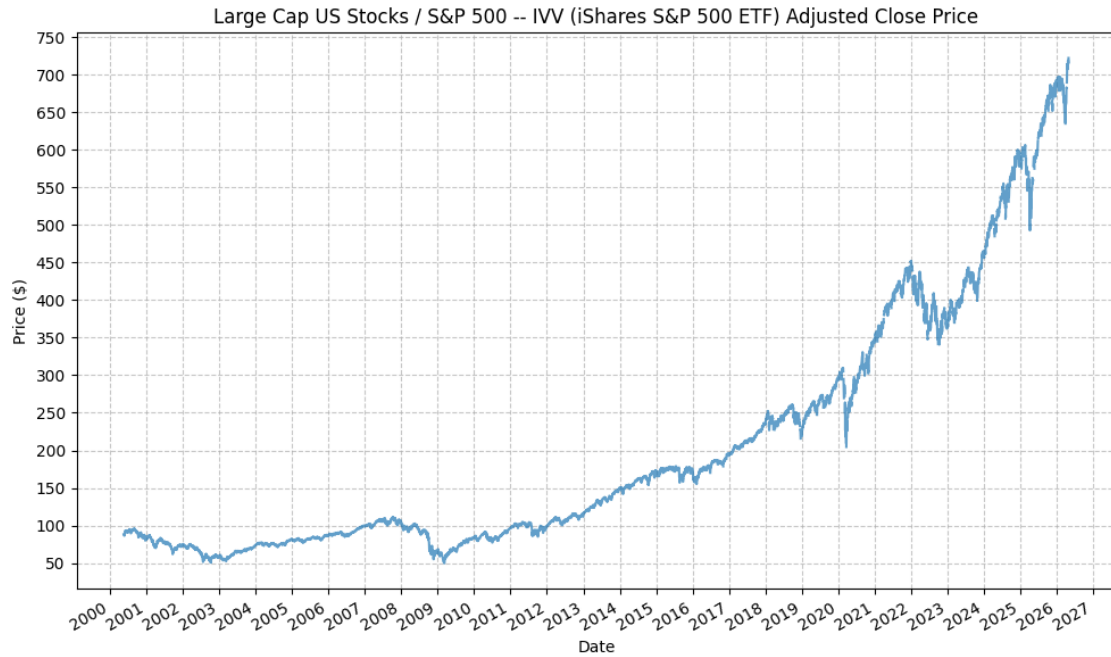
...	...	...
2026-04-27	2303.06	-0.03
2026-04-28	2289.42	-0.01
2026-04-29	2253.42	-0.02
2026-04-30	2256.25	0.00
2026-05-01	2295.09	0.02

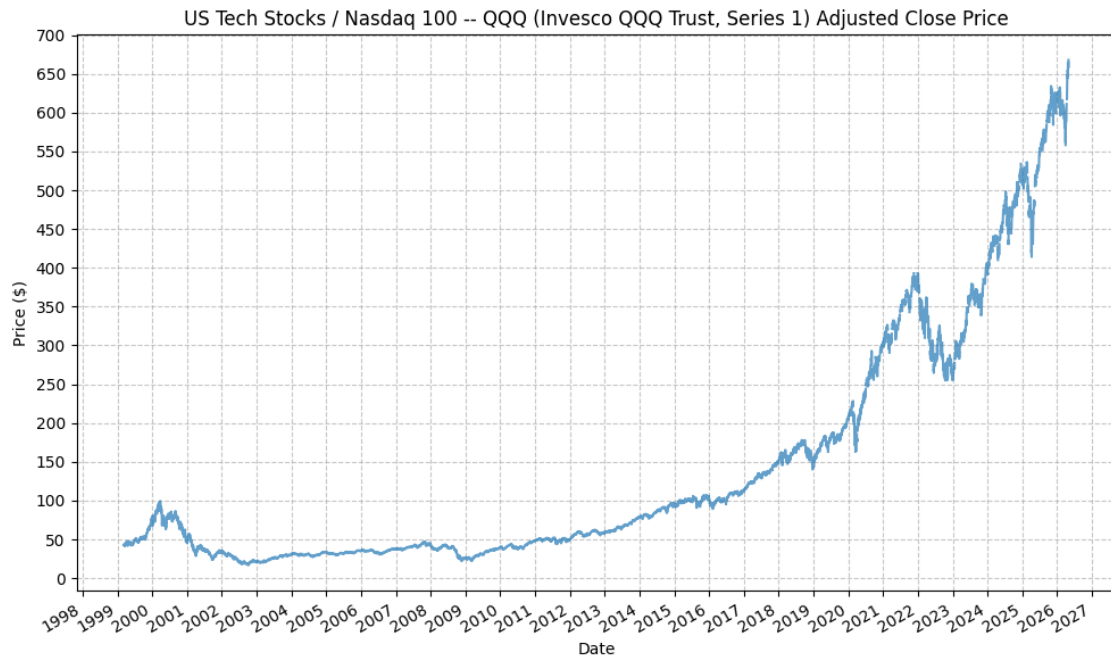
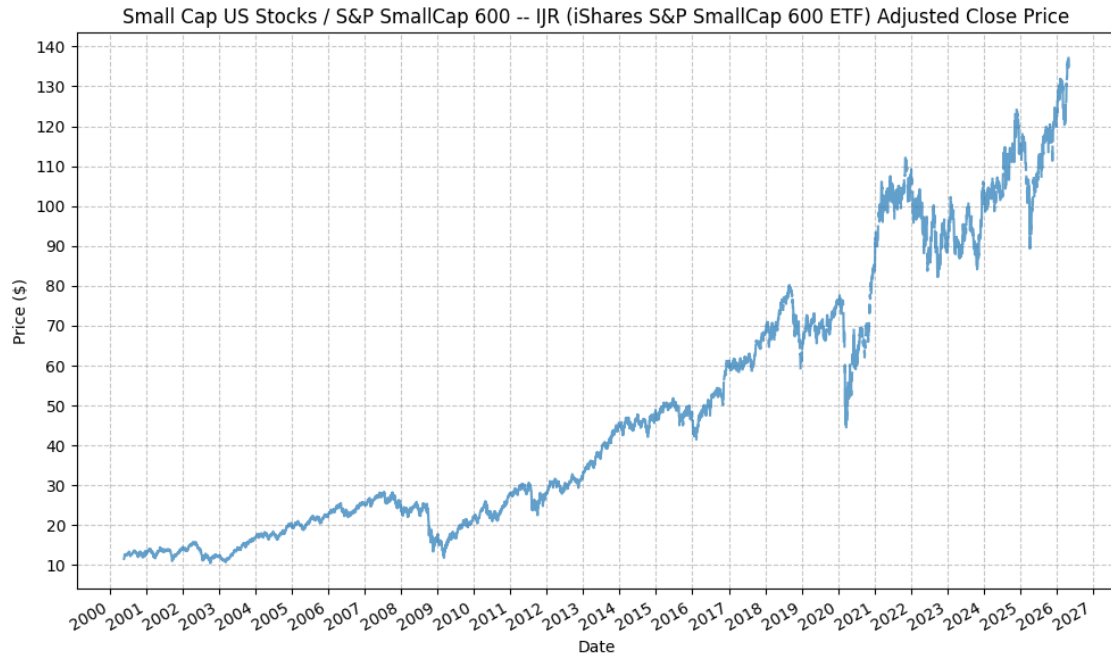
[8151 rows x 40 columns]

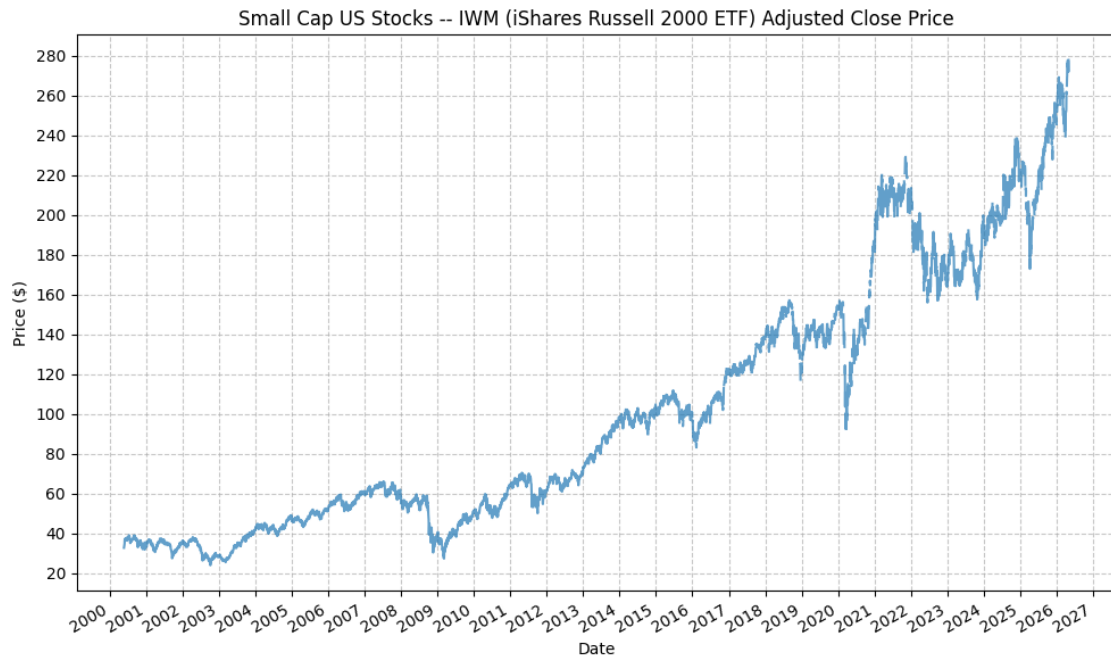
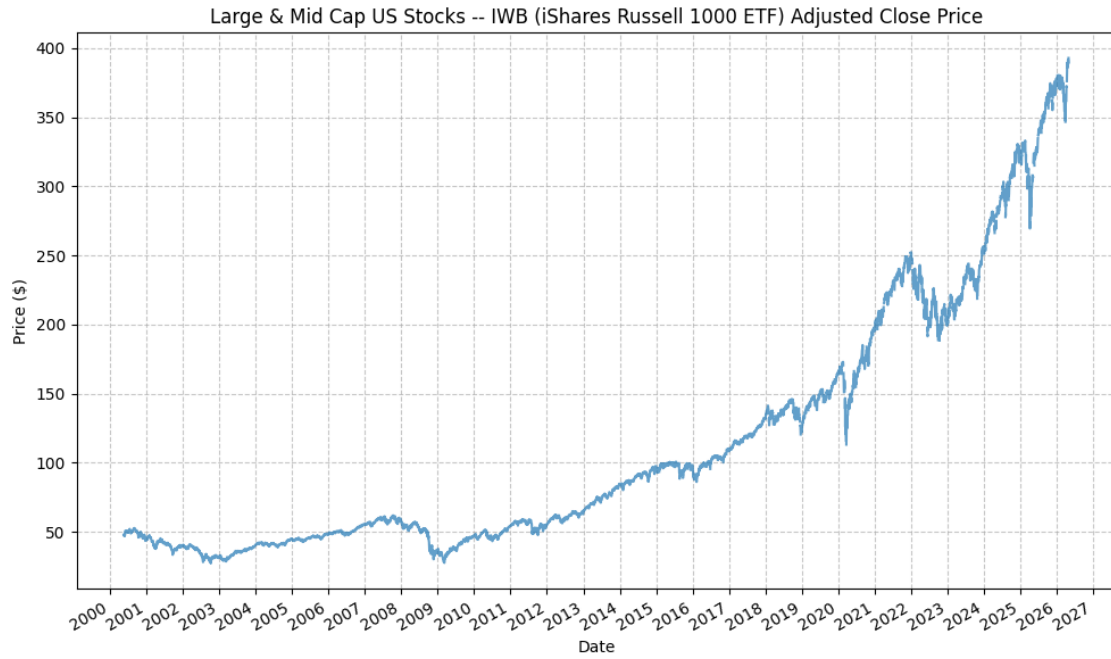
We'll then plot the time series of the adjusted close prices for each of the assets.

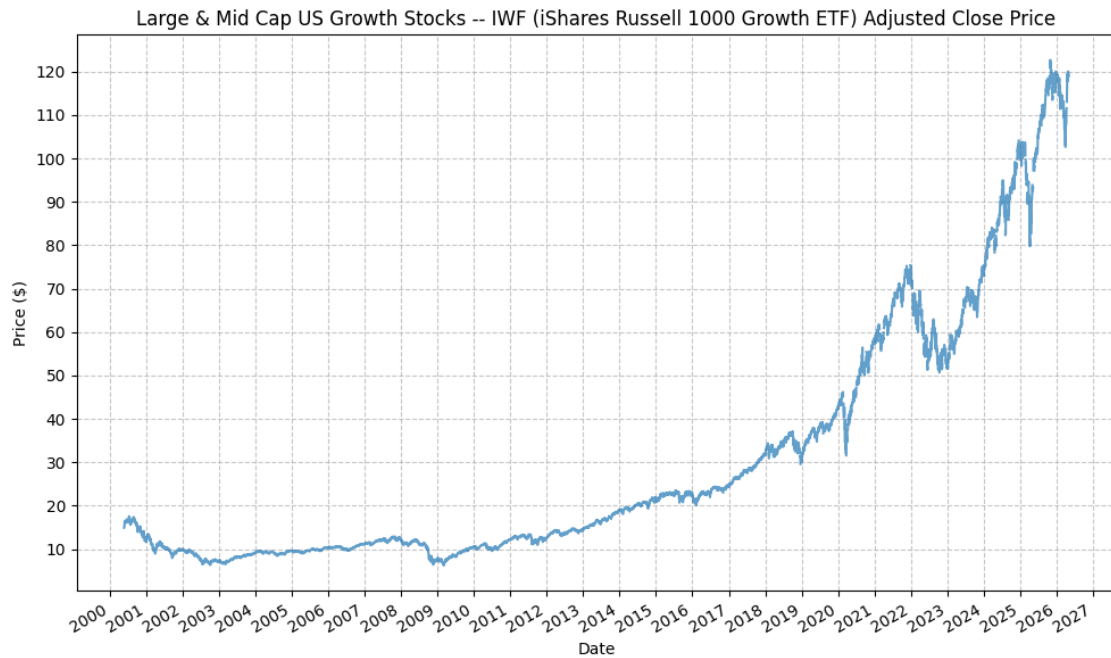
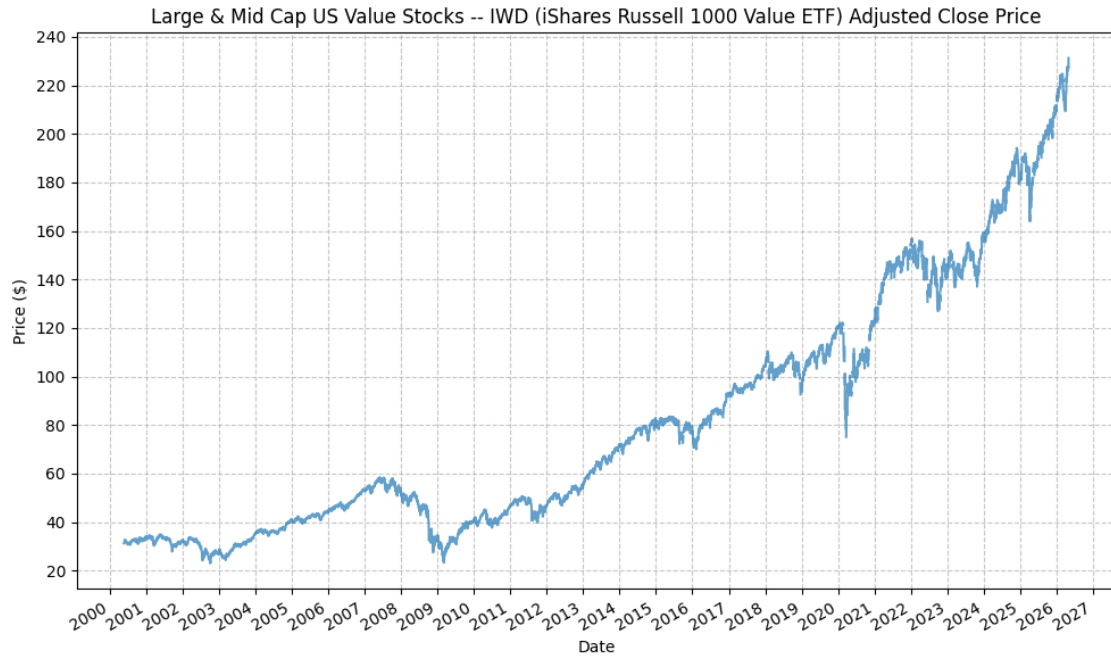
```
[7]: # Combine the etf_tickers and cryptoasset_tickers lists into a single list of
      ↪ all tickers
all_tickers = etf_tickers + cryptoasset_tickers

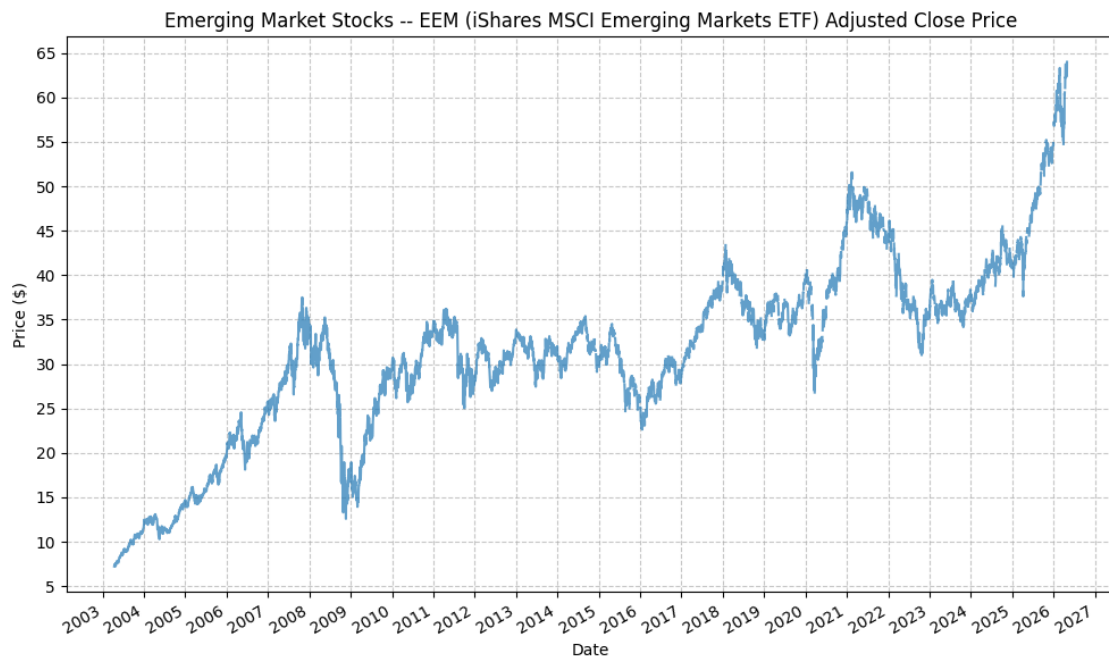
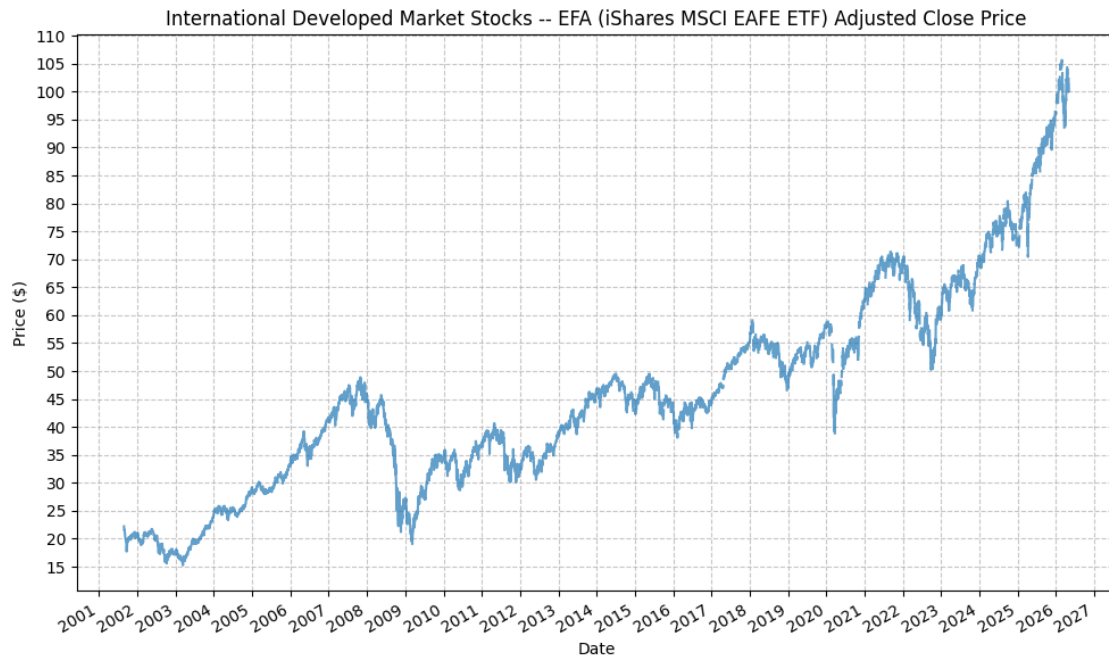
for ticker in tickers_dict.keys():
    plot_time_series(
        df=fund_data,
        plot_start_date=None,
        plot_end_date=None,
        plot_columns=[f"{ticker}_Adj_Close"],
        title=f"{tickers_dict[ticker]} Adjusted Close Price",
        x_label="Date",
        x_format="Year",
        x_tick_spacing=1,
        x_tick_rotation=30,
        y_label="Price ($)",
        y_format="Decimal",
        y_format_decimal_places="Auto",
        y_tick_spacing="Auto",
        y_tick_rotation=0,
        grid=True,
        legend=False,
        export_plot=False,
        plot_file_name=None,
    )
```

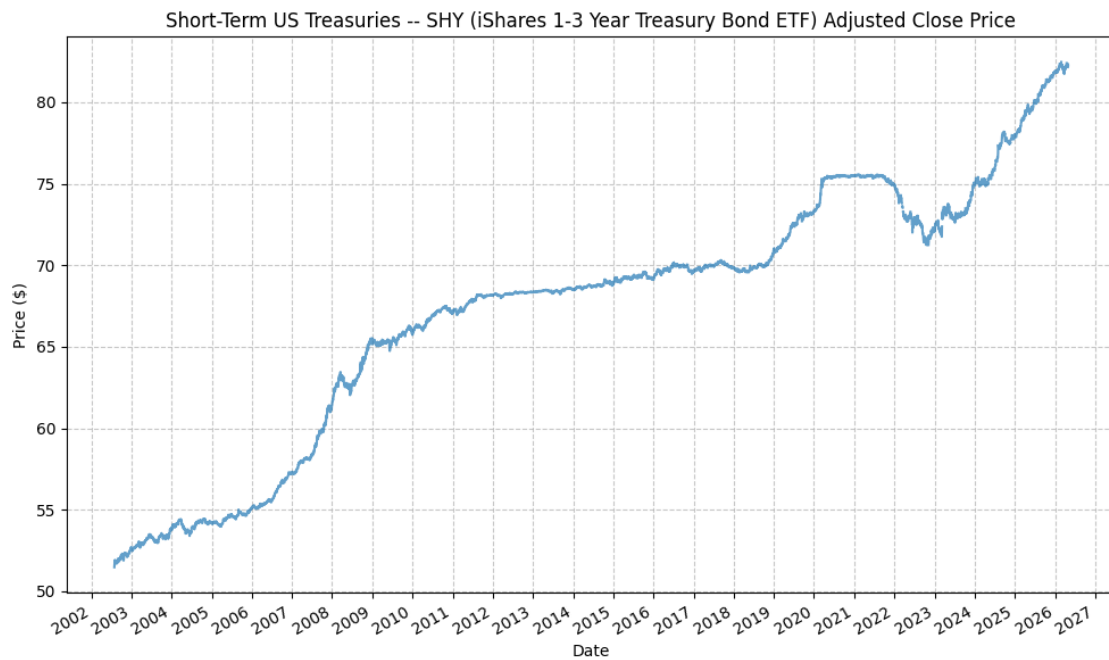
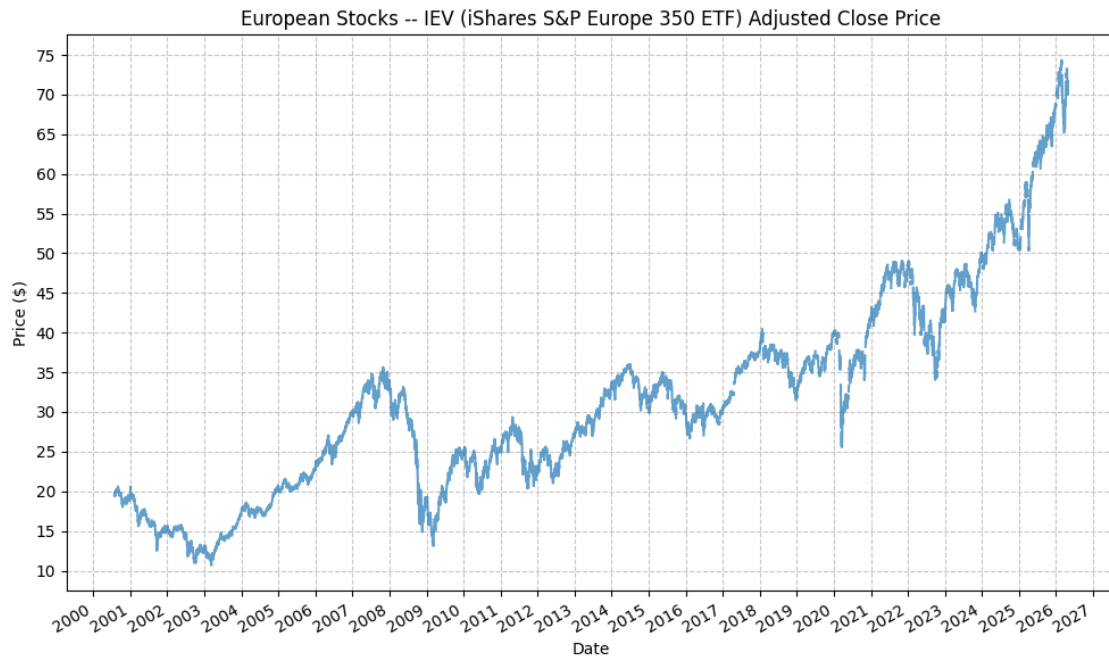



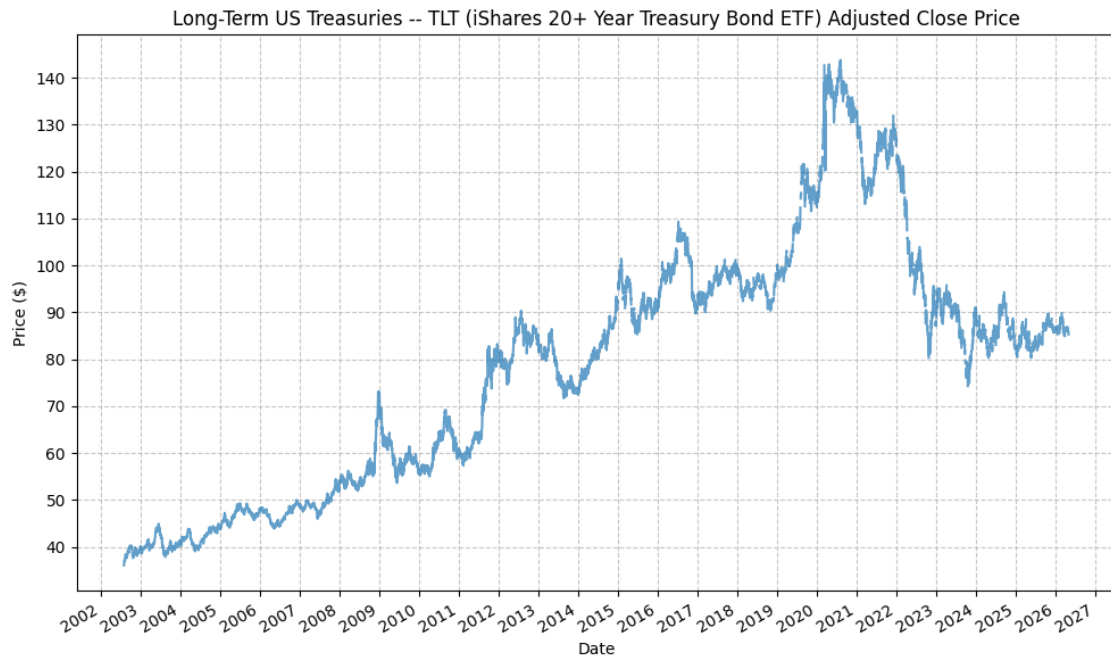
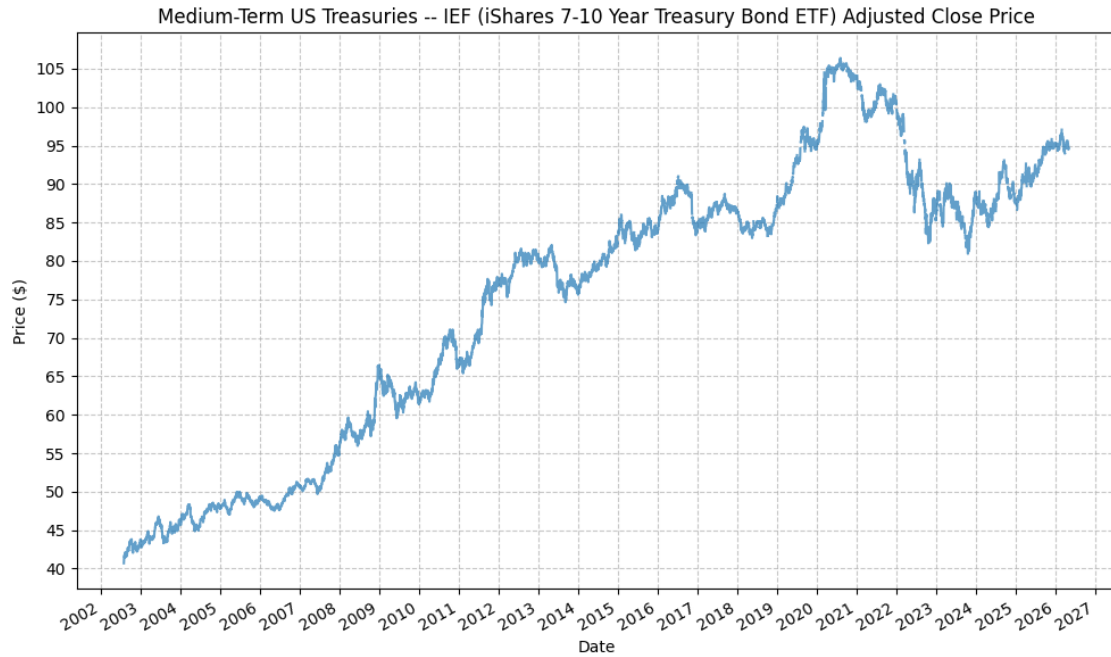


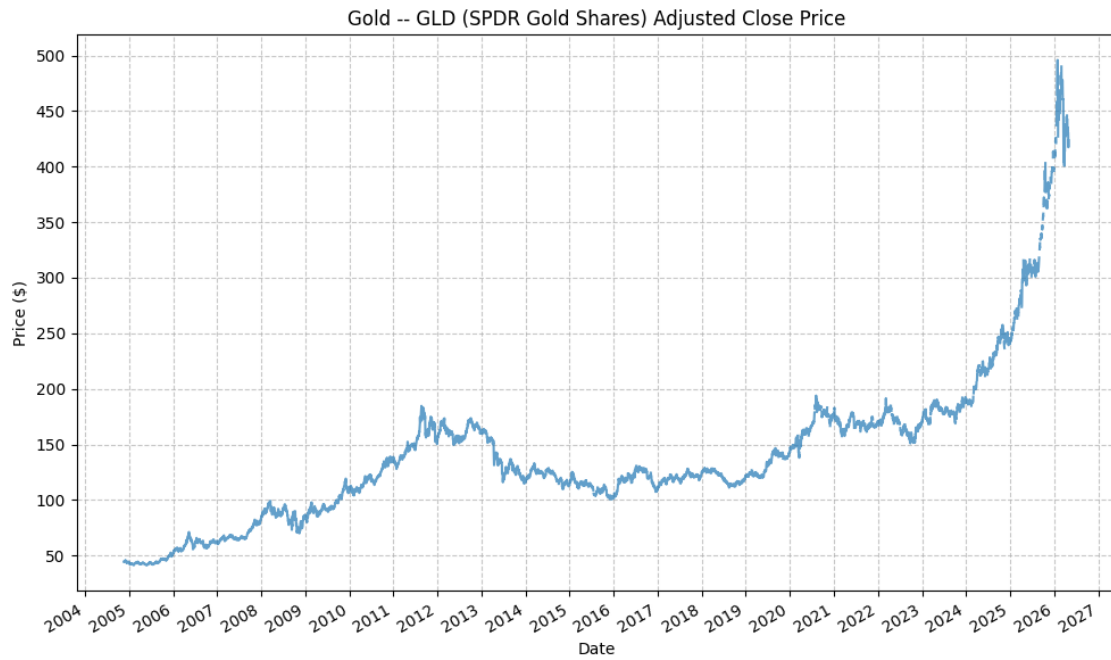
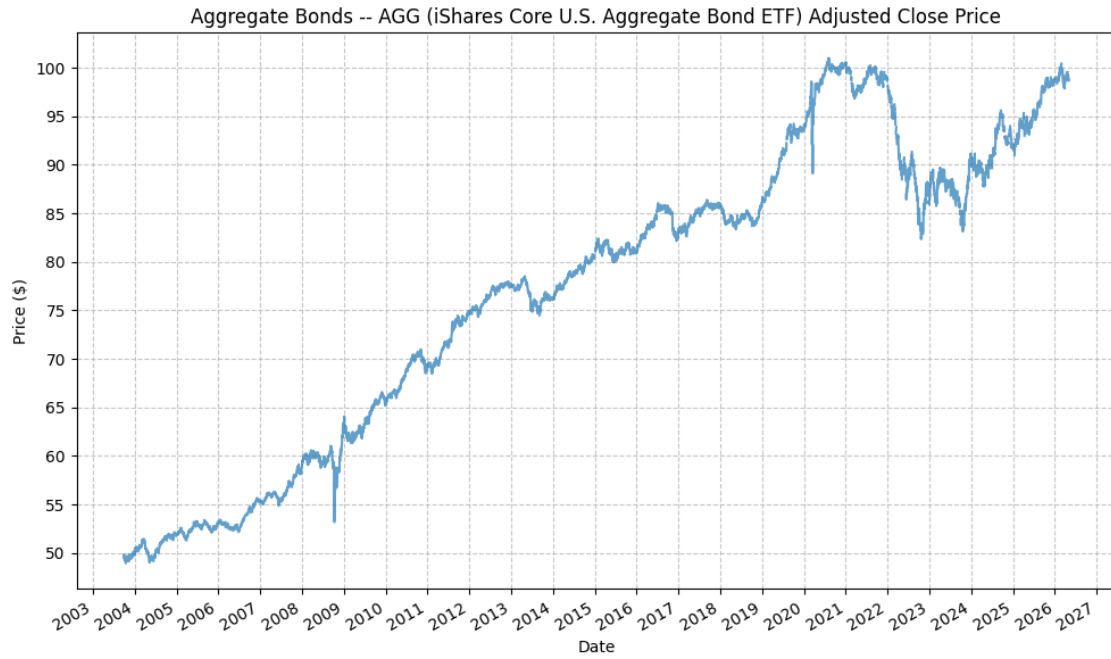


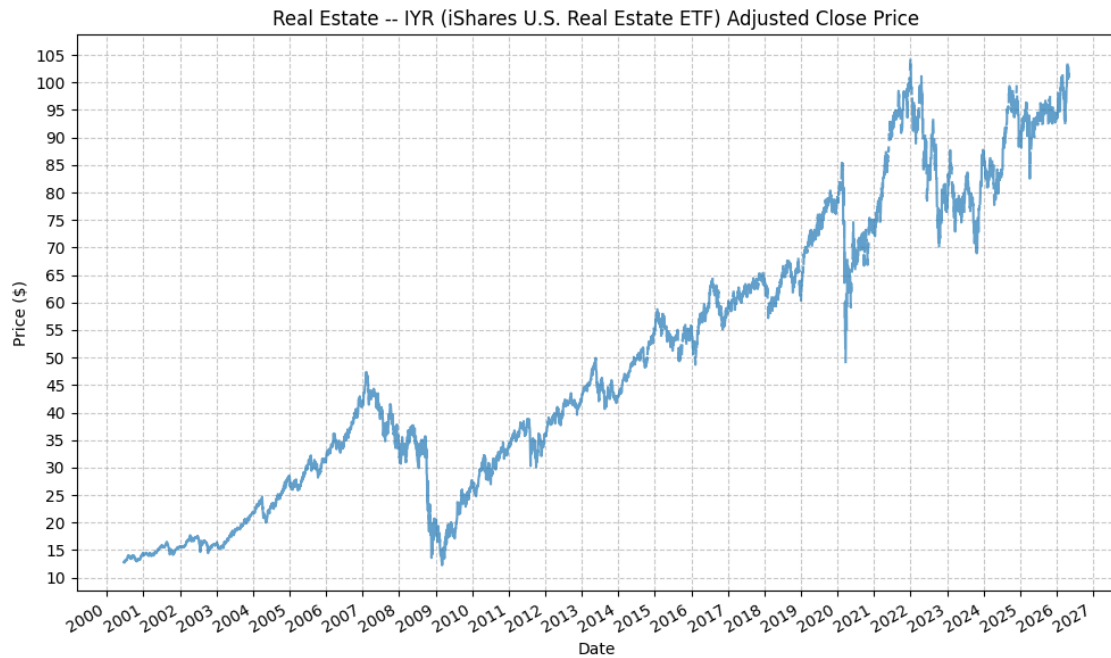
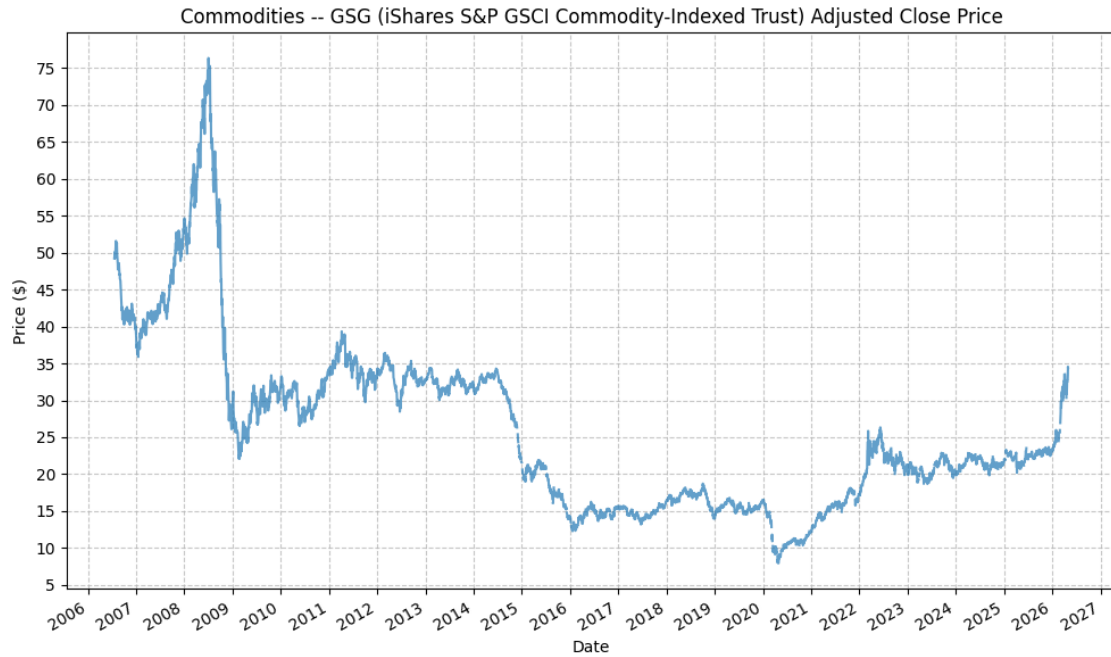


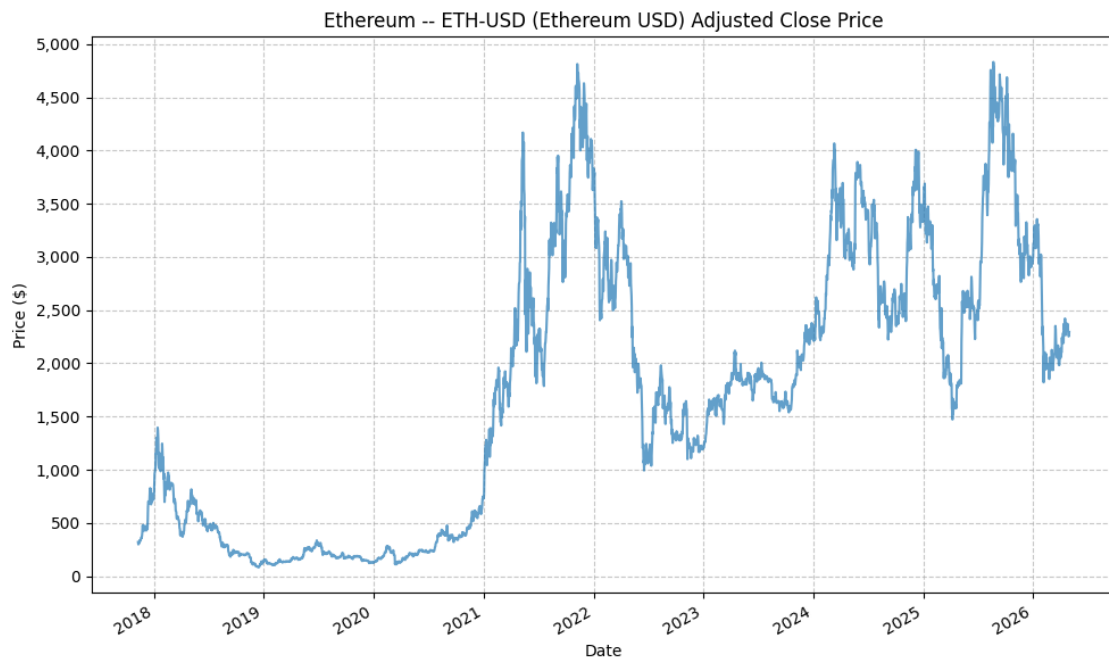
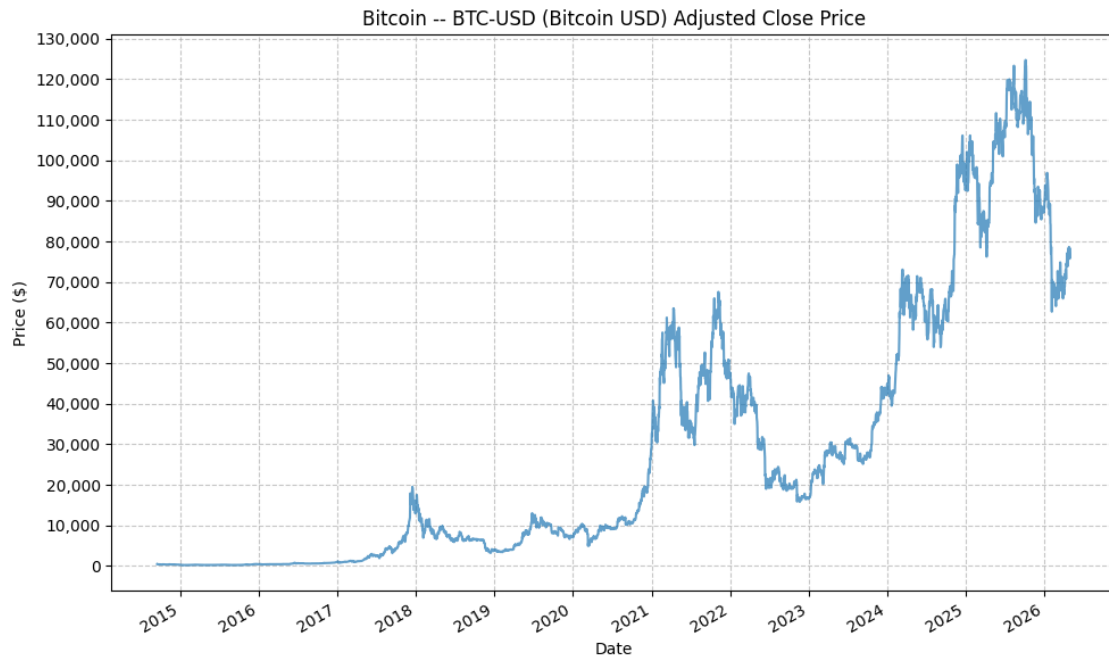












We can see from the plots that there are some gaps in the data, so we'll be sure to drop any rows with missing data before we calculate the correlations.

0.5 Calculate Correlations

Next, we'll calculate the correlation matrix of the daily returns for all of the assets and plot it as a heatmap. We'll do this first without the BTC and ETH data (due to the limited history), and then with the BTC and ETH data.

```
[8]: # Drop the adjusted close price columns, leaving only the daily return columns
daily_return_columns = [f"{ticker}_Daily_Return" for ticker in tickers_dict.
    ↪keys()]
fund_data_daily_returns_all = fund_data[daily_return_columns]

# Drop the BTC and ETH daily return columns due to the limited history
fund_data_daily_returns_no_crypto = fund_data_daily_returns_all.
    ↪drop(columns=["BTC-USD_Daily_Return", "ETH-USD_Daily_Return"])

# Print the shape of the fund_data_daily_returns_no_crypto DataFrame to confirm
    ↪that the rows with missing data have been dropped
print(f"Shape of fund_data_daily_returns_no_crypto:
    ↪{fund_data_daily_returns_no_crypto.shape}")
print(f"Rows to drop due to missing data: {fund_data_daily_returns_no_crypto.
    ↪dropna().shape}")

# Drop the NaN values
fund_data_daily_returns_no_crypto = fund_data_daily_returns_no_crypto.dropna()

# Calculate the correlation matrix of the daily returns
correlation_matrix_no_crypto = fund_data_daily_returns_no_crypto.corr()

display(correlation_matrix_no_crypto)
```

Shape of fund_data_daily_returns_no_crypto: (8151, 18)

Rows to drop due to missing data: (4974, 18)

	IVV_Daily_Return	IJH_Daily_Return	IJR_Daily_Return	\
IVV_Daily_Return	1.00	0.93	0.88	
IJH_Daily_Return	0.93	1.00	0.96	
IJR_Daily_Return	0.88	0.96	1.00	
QQQ_Daily_Return	0.93	0.84	0.79	
IWB_Daily_Return	1.00	0.94	0.89	
IWM_Daily_Return	0.90	0.96	0.98	
IWD_Daily_Return	0.96	0.94	0.90	
IWF_Daily_Return	0.96	0.88	0.82	
EFA_Daily_Return	0.88	0.85	0.80	
EEM_Daily_Return	0.82	0.79	0.74	
IEV_Daily_Return	0.86	0.83	0.78	
SHY_Daily_Return	-0.23	-0.21	-0.20	
IEF_Daily_Return	-0.30	-0.29	-0.28	
TLT_Daily_Return	-0.32	-0.30	-0.30	
AGG_Daily_Return	-0.02	-0.01	-0.04	

GLD_Daily_Return	0.06	0.07	0.04
GSG_Daily_Return	0.38	0.39	0.36
IYR_Daily_Return	0.77	0.79	0.76

	QQQ_Daily_Return	IWB_Daily_Return	IWM_Daily_Return	\
IVV_Daily_Return	0.93	1.00	0.90	
IJH_Daily_Return	0.84	0.94	0.96	
IJR_Daily_Return	0.79	0.89	0.98	
QQQ_Daily_Return	1.00	0.93	0.82	
IWB_Daily_Return	0.93	1.00	0.91	
IWM_Daily_Return	0.82	0.91	1.00	
IWD_Daily_Return	0.82	0.96	0.90	
IWF_Daily_Return	0.97	0.97	0.85	
EFA_Daily_Return	0.79	0.88	0.81	
EEM_Daily_Return	0.76	0.82	0.76	
IEV_Daily_Return	0.76	0.86	0.80	
SHY_Daily_Return	-0.20	-0.23	-0.20	
IEF_Daily_Return	-0.25	-0.30	-0.28	
TLT_Daily_Return	-0.26	-0.31	-0.29	
AGG_Daily_Return	-0.00	-0.01	-0.03	
GLD_Daily_Return	0.05	0.06	0.06	
GSG_Daily_Return	0.30	0.38	0.36	
IYR_Daily_Return	0.65	0.77	0.77	

	IWD_Daily_Return	IWF_Daily_Return	EFA_Daily_Return	\
IVV_Daily_Return	0.96	0.96	0.88	
IJH_Daily_Return	0.94	0.88	0.85	
IJR_Daily_Return	0.90	0.82	0.80	
QQQ_Daily_Return	0.82	0.97	0.79	
IWB_Daily_Return	0.96	0.97	0.88	
IWM_Daily_Return	0.90	0.85	0.81	
IWD_Daily_Return	1.00	0.87	0.88	
IWF_Daily_Return	0.87	1.00	0.83	
EFA_Daily_Return	0.88	0.83	1.00	
EEM_Daily_Return	0.81	0.78	0.88	
IEV_Daily_Return	0.87	0.80	0.98	
SHY_Daily_Return	-0.25	-0.20	-0.19	
IEF_Daily_Return	-0.32	-0.26	-0.27	
TLT_Daily_Return	-0.33	-0.27	-0.29	
AGG_Daily_Return	-0.02	0.02	0.03	
GLD_Daily_Return	0.05	0.06	0.16	
GSG_Daily_Return	0.40	0.34	0.42	
IYR_Daily_Return	0.80	0.68	0.70	

	EEM_Daily_Return	IEV_Daily_Return	SHY_Daily_Return	\
IVV_Daily_Return	0.82	0.86	-0.23	
IJH_Daily_Return	0.79	0.83	-0.21	
IJR_Daily_Return	0.74	0.78	-0.20	

QQQ_Daily_Return	0.76	0.76	-0.20
IWB_Daily_Return	0.82	0.86	-0.23
IWM_Daily_Return	0.76	0.80	-0.20
IWD_Daily_Return	0.81	0.87	-0.25
IWF_Daily_Return	0.78	0.80	-0.20
EFA_Daily_Return	0.88	0.98	-0.19
EEM_Daily_Return	1.00	0.85	-0.22
IEV_Daily_Return	0.85	1.00	-0.19
SHY_Daily_Return	-0.22	-0.19	1.00
IEF_Daily_Return	-0.27	-0.27	0.76
TLT_Daily_Return	-0.27	-0.30	0.57
AGG_Daily_Return	-0.01	0.02	0.60
GLD_Daily_Return	0.18	0.16	0.22
GSG_Daily_Return	0.41	0.41	-0.13
IYR_Daily_Return	0.67	0.69	-0.13

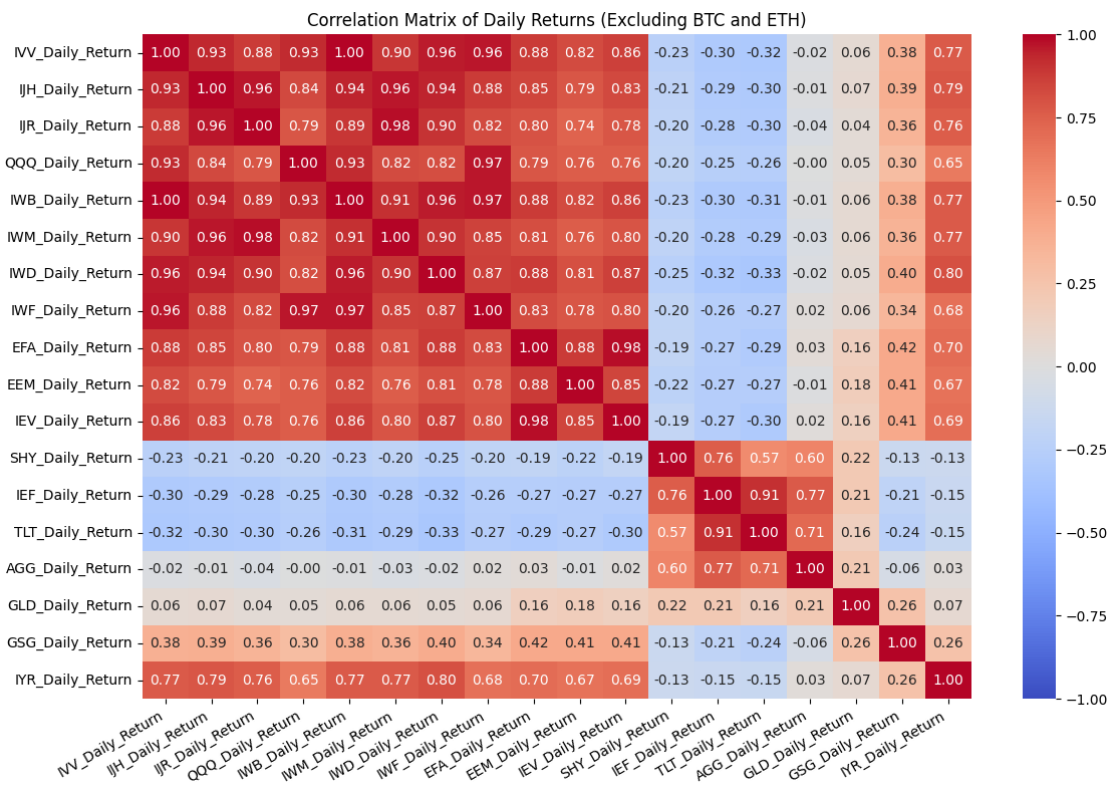
	IEF_Daily_Return	TLT_Daily_Return	AGG_Daily_Return	\
IVV_Daily_Return	-0.30	-0.32	-0.02	
IJH_Daily_Return	-0.29	-0.30	-0.01	
IJR_Daily_Return	-0.28	-0.30	-0.04	
QQQ_Daily_Return	-0.25	-0.26	-0.00	
IWB_Daily_Return	-0.30	-0.31	-0.01	
IWM_Daily_Return	-0.28	-0.29	-0.03	
IWD_Daily_Return	-0.32	-0.33	-0.02	
IWF_Daily_Return	-0.26	-0.27	0.02	
EFA_Daily_Return	-0.27	-0.29	0.03	
EEM_Daily_Return	-0.27	-0.27	-0.01	
IEV_Daily_Return	-0.27	-0.30	0.02	
SHY_Daily_Return	0.76	0.57	0.60	
IEF_Daily_Return	1.00	0.91	0.77	
TLT_Daily_Return	0.91	1.00	0.71	
AGG_Daily_Return	0.77	0.71	1.00	
GLD_Daily_Return	0.21	0.16	0.21	
GSG_Daily_Return	-0.21	-0.24	-0.06	
IYR_Daily_Return	-0.15	-0.15	0.03	

	GLD_Daily_Return	GSG_Daily_Return	IYR_Daily_Return
IVV_Daily_Return	0.06	0.38	0.77
IJH_Daily_Return	0.07	0.39	0.79
IJR_Daily_Return	0.04	0.36	0.76
QQQ_Daily_Return	0.05	0.30	0.65
IWB_Daily_Return	0.06	0.38	0.77
IWM_Daily_Return	0.06	0.36	0.77
IWD_Daily_Return	0.05	0.40	0.80
IWF_Daily_Return	0.06	0.34	0.68
EFA_Daily_Return	0.16	0.42	0.70
EEM_Daily_Return	0.18	0.41	0.67
IEV_Daily_Return	0.16	0.41	0.69

SHY_Daily_Return	0.22	-0.13	-0.13
IEF_Daily_Return	0.21	-0.21	-0.15
TLT_Daily_Return	0.16	-0.24	-0.15
AGG_Daily_Return	0.21	-0.06	0.03
GLD_Daily_Return	1.00	0.26	0.07
GSG_Daily_Return	0.26	1.00	0.26
IYR_Daily_Return	0.07	0.26	1.00

And then the heatmap:

```
[9]: plot_heatmap(
      df=correlation_matrix_no_crypto,
      title="Correlation Matrix of Daily Returns (Excluding BTC and ETH)",
    )
```



We'll now include the BTC and ETH data and recalculate the correlation matrix and heatmap. Keep in mind that the BTC and ETH data only goes back to 2015 and 2018 respectively, so the correlation matrix will be calculated using a shorter time period than the other assets.

```
[10]: # Print the shape of the fund_data_daily_returns DataFrame to confirm that the
      ↪ rows with missing data have been dropped
      print(f"Shape of fund_data_daily_returns: {fund_data_daily_returns_all.shape}")
```

```

print(f"Rows to drop due to missing data: {fund_data_daily_returns_all.dropna().
↪shape}")

# Drop the NaN values
fund_data_daily_returns = fund_data_daily_returns_all.dropna()

# Calculate the correlation matrix of the daily returns
correlation_matrix = fund_data_daily_returns.corr()

display(correlation_matrix)

```

Shape of fund_data_daily_returns: (8151, 20)

Rows to drop due to missing data: (2127, 20)

	IVV_Daily_Return	IJH_Daily_Return	IJR_Daily_Return	\
IVV_Daily_Return	1.00	0.90	0.84	
IJH_Daily_Return	0.90	1.00	0.96	
IJR_Daily_Return	0.84	0.96	1.00	
QQQ_Daily_Return	0.94	0.78	0.71	
IWB_Daily_Return	1.00	0.91	0.85	
IWM_Daily_Return	0.87	0.97	0.98	
IWD_Daily_Return	0.93	0.94	0.90	
IWF_Daily_Return	0.96	0.81	0.74	
EFA_Daily_Return	0.85	0.84	0.79	
EEM_Daily_Return	0.76	0.72	0.67	
IEV_Daily_Return	0.82	0.81	0.76	
SHY_Daily_Return	-0.06	-0.05	-0.04	
IEF_Daily_Return	-0.12	-0.12	-0.10	
TLT_Daily_Return	-0.15	-0.14	-0.13	
AGG_Daily_Return	0.14	0.15	0.15	
GLD_Daily_Return	0.10	0.10	0.08	
GSG_Daily_Return	0.31	0.32	0.31	
IYR_Daily_Return	0.74	0.78	0.74	
BTC-USD_Daily_Return	0.30	0.29	0.28	
ETH-USD_Daily_Return	0.33	0.31	0.30	

	QQQ_Daily_Return	IWB_Daily_Return	IWM_Daily_Return	\
IVV_Daily_Return	0.94	1.00	0.87	
IJH_Daily_Return	0.78	0.91	0.97	
IJR_Daily_Return	0.71	0.85	0.98	
QQQ_Daily_Return	1.00	0.94	0.77	
IWB_Daily_Return	0.94	1.00	0.88	
IWM_Daily_Return	0.77	0.88	1.00	
IWD_Daily_Return	0.76	0.93	0.89	
IWF_Daily_Return	0.99	0.96	0.80	
EFA_Daily_Return	0.76	0.85	0.80	
EEM_Daily_Return	0.73	0.76	0.71	
IEV_Daily_Return	0.72	0.83	0.78	

SHY_Daily_Return	-0.03	-0.05	-0.03
IEF_Daily_Return	-0.08	-0.12	-0.09
TLT_Daily_Return	-0.10	-0.14	-0.12
AGG_Daily_Return	0.14	0.14	0.16
GLD_Daily_Return	0.11	0.10	0.11
GSG_Daily_Return	0.24	0.31	0.31
IYR_Daily_Return	0.59	0.75	0.73
BTC-USD_Daily_Return	0.31	0.30	0.31
ETH-USD_Daily_Return	0.34	0.33	0.33

	IWD_Daily_Return	IWF_Daily_Return	EFA_Daily_Return	\
IVV_Daily_Return	0.93	0.96	0.85	
IJH_Daily_Return	0.94	0.81	0.84	
IJR_Daily_Return	0.90	0.74	0.79	
QQQ_Daily_Return	0.76	0.99	0.76	
IWB_Daily_Return	0.93	0.96	0.85	
IWM_Daily_Return	0.89	0.80	0.80	
IWD_Daily_Return	1.00	0.80	0.85	
IWF_Daily_Return	0.80	1.00	0.78	
EFA_Daily_Return	0.85	0.78	1.00	
EEM_Daily_Return	0.71	0.73	0.83	
IEV_Daily_Return	0.83	0.75	0.98	
SHY_Daily_Return	-0.06	-0.03	0.01	
IEF_Daily_Return	-0.14	-0.08	-0.05	
TLT_Daily_Return	-0.18	-0.10	-0.10	
AGG_Daily_Return	0.12	0.15	0.20	
GLD_Daily_Return	0.10	0.10	0.23	
GSG_Daily_Return	0.35	0.25	0.30	
IYR_Daily_Return	0.81	0.64	0.69	
BTC-USD_Daily_Return	0.26	0.31	0.29	
ETH-USD_Daily_Return	0.28	0.34	0.32	

	EEM_Daily_Return	IEV_Daily_Return	SHY_Daily_Return	\
IVV_Daily_Return	0.76	0.82	-0.06	
IJH_Daily_Return	0.72	0.81	-0.05	
IJR_Daily_Return	0.67	0.76	-0.04	
QQQ_Daily_Return	0.73	0.72	-0.03	
IWB_Daily_Return	0.76	0.83	-0.05	
IWM_Daily_Return	0.71	0.78	-0.03	
IWD_Daily_Return	0.71	0.83	-0.06	
IWF_Daily_Return	0.73	0.75	-0.03	
EFA_Daily_Return	0.83	0.98	0.01	
EEM_Daily_Return	1.00	0.80	-0.01	
IEV_Daily_Return	0.80	1.00	0.01	
SHY_Daily_Return	-0.01	0.01	1.00	
IEF_Daily_Return	-0.07	-0.06	0.81	
TLT_Daily_Return	-0.10	-0.10	0.60	
AGG_Daily_Return	0.15	0.20	0.73	

GLD_Daily_Return	0.25	0.22	0.31
GSG_Daily_Return	0.30	0.29	-0.12
IYR_Daily_Return	0.56	0.68	0.13
BTC-USD_Daily_Return	0.27	0.29	0.03
ETH-USD_Daily_Return	0.31	0.31	0.02

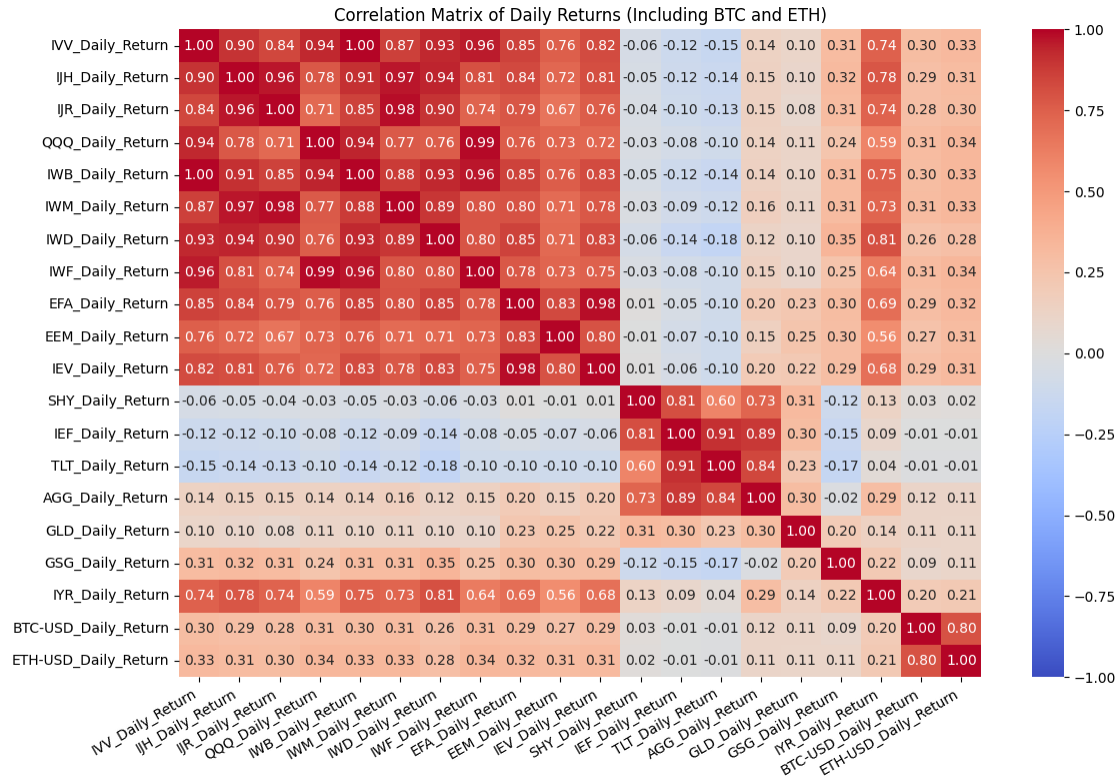
	IEF_Daily_Return	TLT_Daily_Return	AGG_Daily_Return \
IVV_Daily_Return	-0.12	-0.15	0.14
IJH_Daily_Return	-0.12	-0.14	0.15
IJR_Daily_Return	-0.10	-0.13	0.15
QQQ_Daily_Return	-0.08	-0.10	0.14
IWB_Daily_Return	-0.12	-0.14	0.14
IWM_Daily_Return	-0.09	-0.12	0.16
IWD_Daily_Return	-0.14	-0.18	0.12
IWF_Daily_Return	-0.08	-0.10	0.15
EFA_Daily_Return	-0.05	-0.10	0.20
EEM_Daily_Return	-0.07	-0.10	0.15
IEV_Daily_Return	-0.06	-0.10	0.20
SHY_Daily_Return	0.81	0.60	0.73
IEF_Daily_Return	1.00	0.91	0.89
TLT_Daily_Return	0.91	1.00	0.84
AGG_Daily_Return	0.89	0.84	1.00
GLD_Daily_Return	0.30	0.23	0.30
GSG_Daily_Return	-0.15	-0.17	-0.02
IYR_Daily_Return	0.09	0.04	0.29
BTC-USD_Daily_Return	-0.01	-0.01	0.12
ETH-USD_Daily_Return	-0.01	-0.01	0.11

	GLD_Daily_Return	GSG_Daily_Return	IYR_Daily_Return \
IVV_Daily_Return	0.10	0.31	0.74
IJH_Daily_Return	0.10	0.32	0.78
IJR_Daily_Return	0.08	0.31	0.74
QQQ_Daily_Return	0.11	0.24	0.59
IWB_Daily_Return	0.10	0.31	0.75
IWM_Daily_Return	0.11	0.31	0.73
IWD_Daily_Return	0.10	0.35	0.81
IWF_Daily_Return	0.10	0.25	0.64
EFA_Daily_Return	0.23	0.30	0.69
EEM_Daily_Return	0.25	0.30	0.56
IEV_Daily_Return	0.22	0.29	0.68
SHY_Daily_Return	0.31	-0.12	0.13
IEF_Daily_Return	0.30	-0.15	0.09
TLT_Daily_Return	0.23	-0.17	0.04
AGG_Daily_Return	0.30	-0.02	0.29
GLD_Daily_Return	1.00	0.20	0.14
GSG_Daily_Return	0.20	1.00	0.22
IYR_Daily_Return	0.14	0.22	1.00
BTC-USD_Daily_Return	0.11	0.09	0.20

ETH-USD_Daily_Return	0.11	0.11	0.21
	BTC-USD_Daily_Return	ETH-USD_Daily_Return	
IVV_Daily_Return	0.30	0.33	
IJH_Daily_Return	0.29	0.31	
IJR_Daily_Return	0.28	0.30	
QQQ_Daily_Return	0.31	0.34	
IWB_Daily_Return	0.30	0.33	
IWM_Daily_Return	0.31	0.33	
IWD_Daily_Return	0.26	0.28	
IWF_Daily_Return	0.31	0.34	
EFA_Daily_Return	0.29	0.32	
EEM_Daily_Return	0.27	0.31	
IEV_Daily_Return	0.29	0.31	
SHY_Daily_Return	0.03	0.02	
IEF_Daily_Return	-0.01	-0.01	
TLT_Daily_Return	-0.01	-0.01	
AGG_Daily_Return	0.12	0.11	
GLD_Daily_Return	0.11	0.11	
GSG_Daily_Return	0.09	0.11	
IYR_Daily_Return	0.20	0.21	
BTC-USD_Daily_Return	1.00	0.80	
ETH-USD_Daily_Return	0.80	1.00	

And then the heatmap:

```
[11]: plot_heatmap(
        df=correlation_matrix,
        title="Correlation Matrix of Daily Returns (Including BTC and ETH)",
    )
```



These are interesting results, but expected. The stock funds tend to have low correlations with bond funds, the commodities don't really correlate with anything, etc. But we know that the correlations between these asset classes have changed over time, so let's take a look at how the correlations have evolved over time by calculating rolling correlations.

Calculate Rolling Correlations

Next, we will calculate the rolling correlations for several different periods: * 1 month * 3 month * 6 month * 1 year * 5 years * 10 years

```
[12]: # Define rolling windows in trading days
rolling_windows = {
    '1mo': 21,      # 1 month (~21 trading days)
    '3mo': 63,      # 3 months (~63 trading days)
    '6mo': 126,     # 6 months (~126 trading days)
    '1y': 252,      # 1 year (~252 trading days)
    '5y': 1260,     # 5 years (~1260 trading days)
    '10y': 2520,    # 10 years (~2520 trading days)
}
```

Before we run all of these, let's take a quick look at each of the rolling windows to see if they are valid.

```

[13]: temp_df = fund_data_daily_returns_all[["IVV_Daily_Return", "IJH_Daily_Return"]].
      ↪dropna()

for window_name, window_size in rolling_windows.items():
    rolling_corr = temp_df["IVV_Daily_Return"].rolling(window=window_size).
    ↪corr(temp_df["IJH_Daily_Return"])

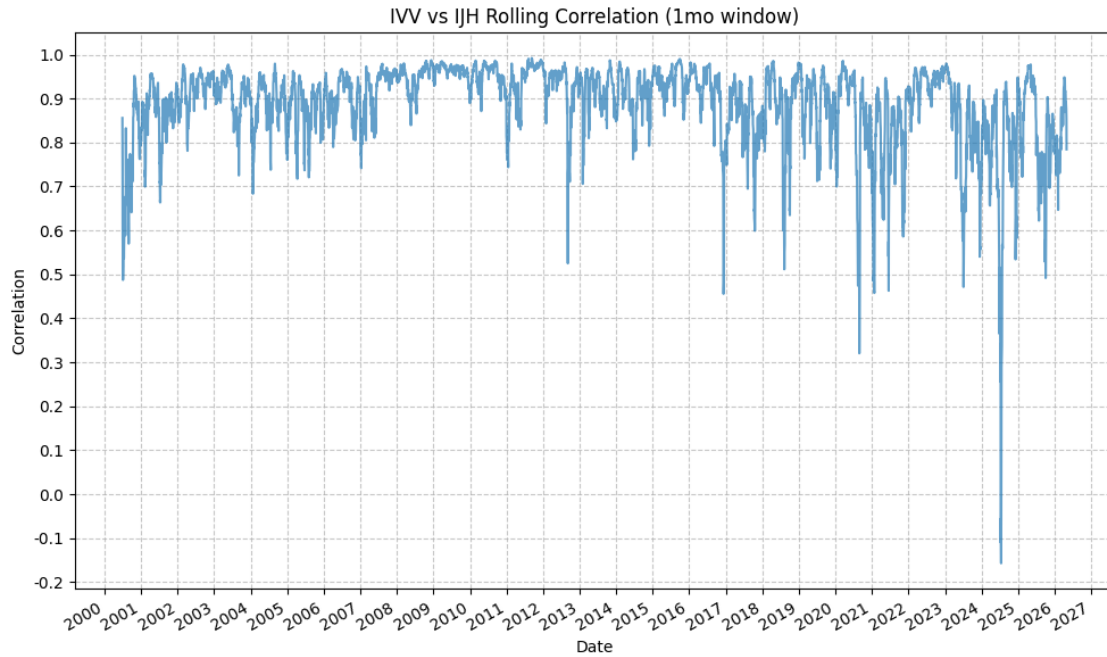
    print(tickers_dict["IVV"])
    print(tickers_dict["IJH"])

    plot_time_series(
        df=rolling_corr.to_frame(name="IVV_IJH_Rolling_Correlation"),
        plot_start_date=None,
        plot_end_date=None,
        plot_columns=["IVV_IJH_Rolling_Correlation"],
        title=f"IVV vs IJH Rolling Correlation ({window_name} window)",
        x_label="Date",
        x_format="Year",
        x_tick_spacing=1,
        x_tick_rotation=30,
        y_label="Correlation",
        y_format="Decimal",
        y_format_decimal_places="Auto",
        y_tick_spacing="Auto",
        y_tick_rotation=0,
        grid=True,
        legend=False,
        export_plot=False,
        plot_file_name=None,
    )

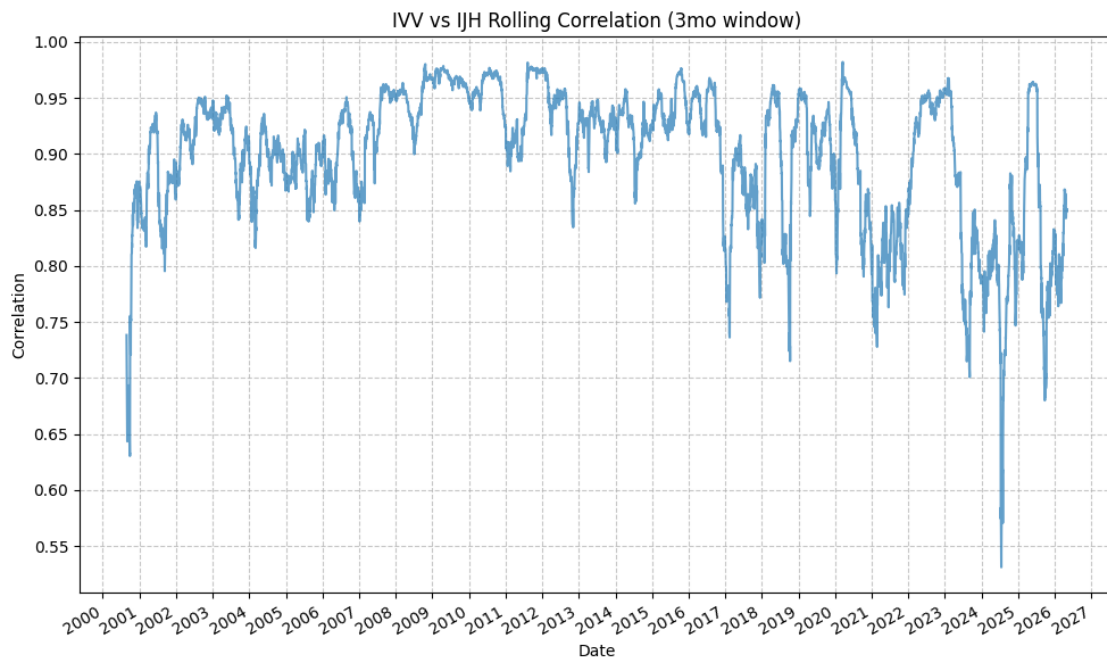
```

Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)

Mid Cap US Stocks / S&P MidCap 400 -- IJH (iShares S&P MidCap 400 ETF)

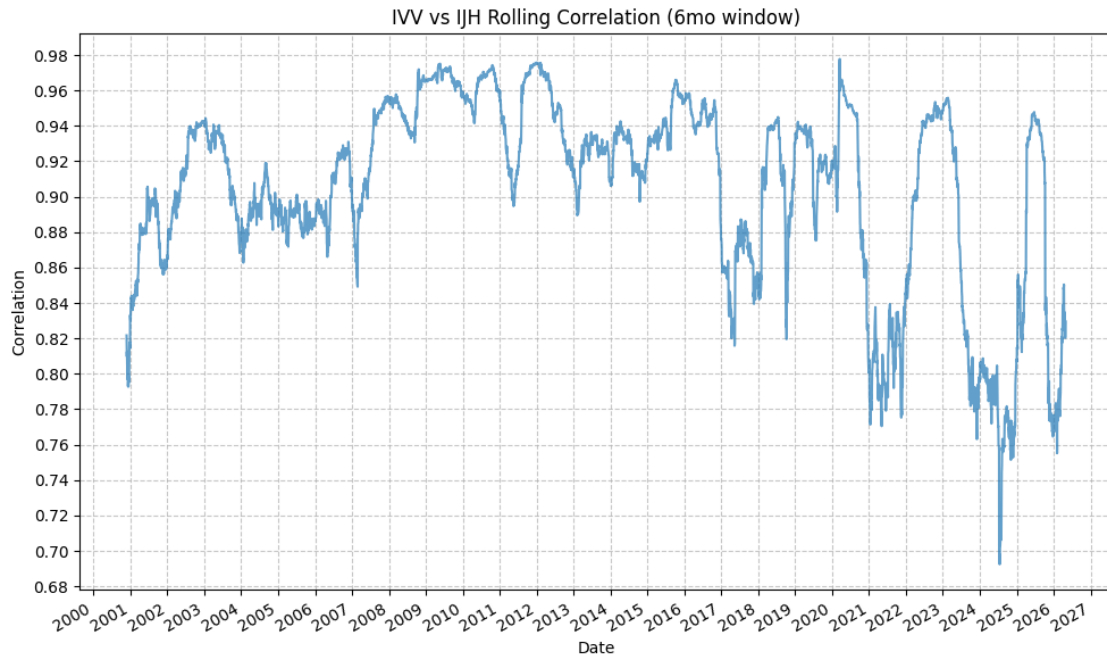


Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)
 Mid Cap US Stocks / S&P MidCap 400 -- IJH (iShares S&P MidCap 400 ETF)



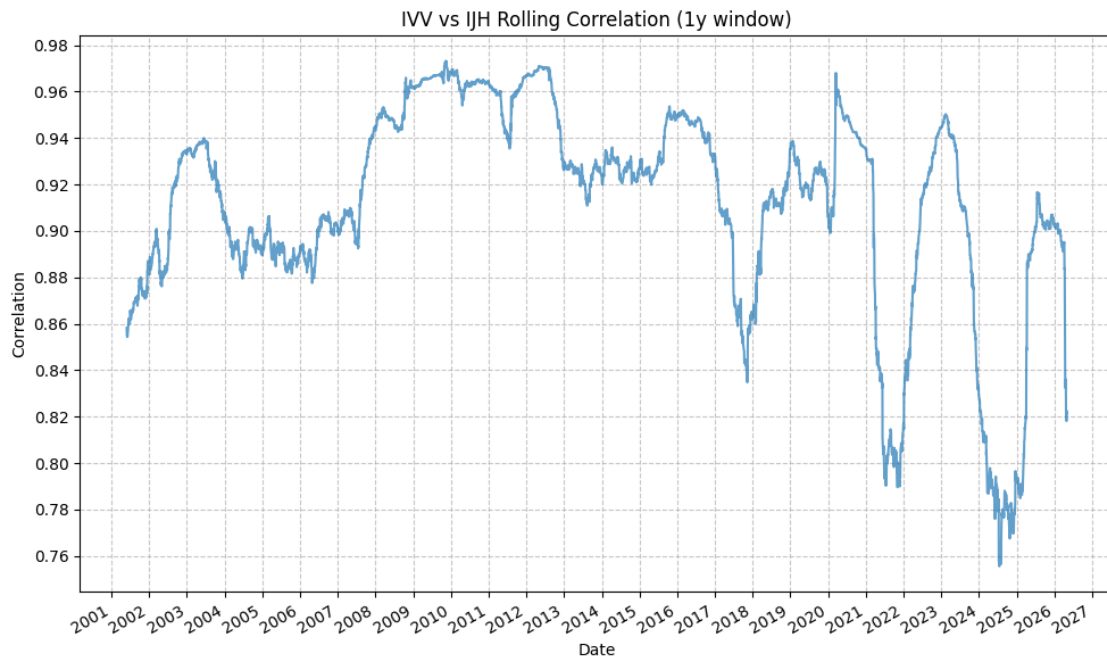
Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)

Mid Cap US Stocks / S&P MidCap 400 -- IJH (iShares S&P MidCap 400 ETF)



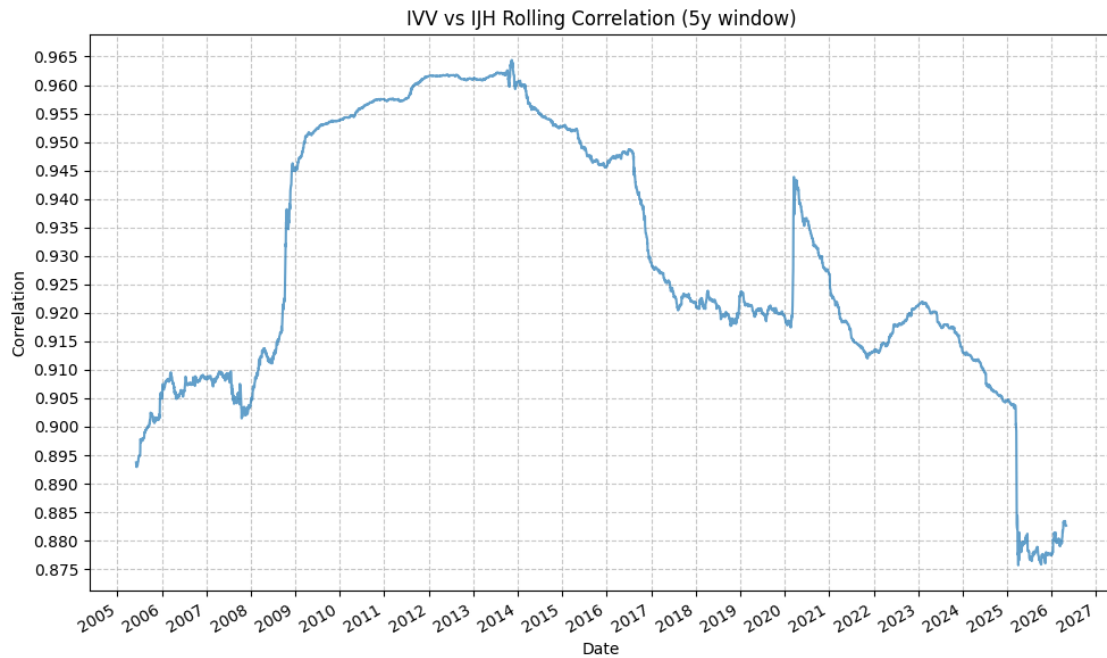
Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)

Mid Cap US Stocks / S&P MidCap 400 -- IJH (iShares S&P MidCap 400 ETF)



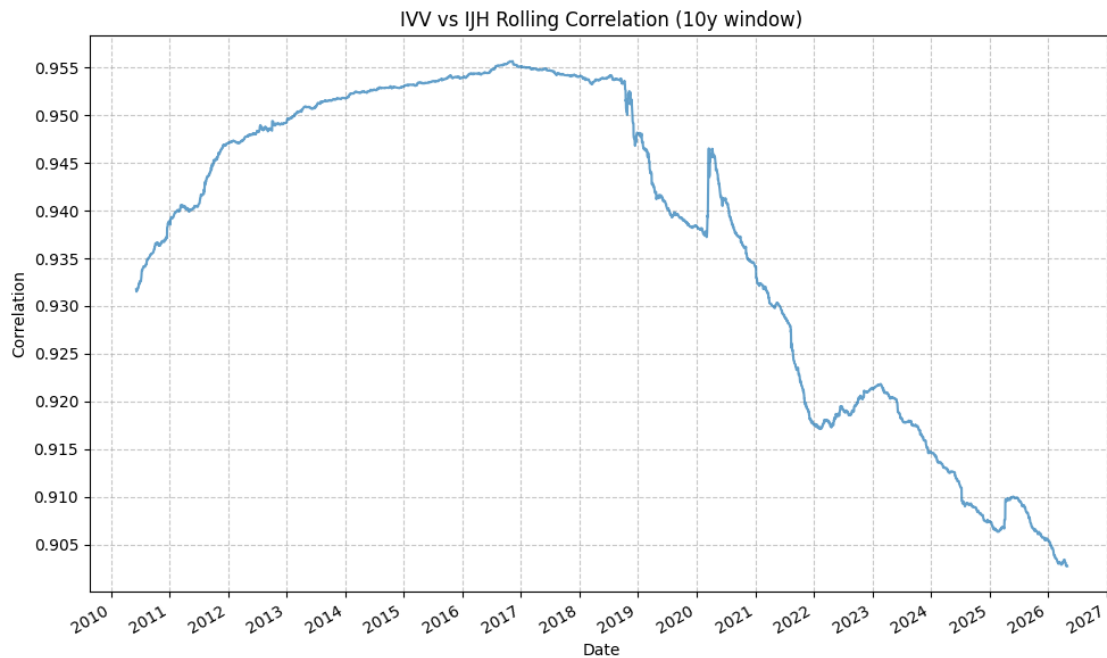
Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)

Mid Cap US Stocks / S&P MidCap 400 -- IJH (iShares S&P MidCap 400 ETF)



Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)

Mid Cap US Stocks / S&P MidCap 400 -- IJH (iShares S&P MidCap 400 ETF)



Some initial thoughts here: * The 1 month and 3 month rolling windows are very noisy... it's going to be difficult to capture any kinds of meaningful trends with these short rolling windows - and these short-term movements are not the point of our investigation here. * The 6 months and 1 year rolling windows look like they might be a bit better, but still pretty noisy. We'll keep both of those. * The 5 year also looks potentially useful, but the 10 year is too long - it doesn't capture any of the medium-term movements that we are interested in. We'll keep the 5 year and drop the 10 year.

So that leaves us with: * 6 months * 1 year * 5 years

We are essentially looking to capture market movements over the months-to-years time frame (from a macro perspective), so the 6 month, 1 year, and 5 year rolling windows are the most appropriate for this analysis.

```
[14]: # Define rolling windows in trading days
rolling_windows = {
    '1mo': 21,      # 1 month (~21 trading days)
    '3mo': 63,      # 3 months (~63 trading days)
    '6mo': 126,     # 6 months (~126 trading days)
    '1y': 252,      # 1 year (~252 trading days)
    '5y': 1260,     # 5 years (~1260 trading days)
}

[15]: from itertools import combinations

# Create temp list for tickers
temp_tickers = list(tickers_dict.keys())
pairs = list(combinations(temp_tickers, 2))

# Create empty dictionary to store rolling correlation results
rolling_correlation_results_no_crypto_dict = {}
rolling_correlation_results_no_crypto_df = pd.DataFrame()

for ticker1, ticker2 in pairs:
    try:
        temp_df = fund_data_daily_returns_no_crypto[[f"{ticker1}_Daily_Return",
↵f"{ticker2}_Daily_Return"]].dropna()
    except Exception as e:
        print(f"Error creating temp_df for {ticker1} and {ticker2}: {e}")
        continue

    for window_name, window_size in rolling_windows.items():
        try:
            rolling_corr = temp_df[f"{ticker1}_Daily_Return"].
↵rolling(window=window_size).corr(temp_df[f"{ticker2}_Daily_Return"])

            print(tickers_dict[f"{ticker1}"])
```



```

        print(tickers_dict[f"{ticker2}"])

    ↵
    ↪rolling_correlation_results_no_crypto_dict[f"{ticker1}_{ticker2}_{window_name}"]↵
    ↪= rolling_corr
        rolling_correlation_results_no_crypto_df = pd.
    ↪concat([rolling_correlation_results_no_crypto_df, rolling_corr.
    ↪to_frame(name=f"{ticker1}_{ticker2}_{window_name}")), axis=1)

    except Exception as e:
        print(f"Error calculating rolling correlation for {ticker1} and↵
    ↪{ticker2} with window {window_name}: {e}")

```

```

Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)
Mid Cap US Stocks / S&P MidCap 400 -- IJH (iShares S&P MidCap 400 ETF)
Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)
Mid Cap US Stocks / S&P MidCap 400 -- IJH (iShares S&P MidCap 400 ETF)
Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)
Mid Cap US Stocks / S&P MidCap 400 -- IJH (iShares S&P MidCap 400 ETF)
Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)
Mid Cap US Stocks / S&P MidCap 400 -- IJH (iShares S&P MidCap 400 ETF)
Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)
Mid Cap US Stocks / S&P MidCap 400 -- IJH (iShares S&P MidCap 400 ETF)
Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)
Small Cap US Stocks / S&P SmallCap 600 -- IJR (iShares S&P SmallCap 600 ETF)
Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)
Small Cap US Stocks / S&P SmallCap 600 -- IJR (iShares S&P SmallCap 600 ETF)
Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)
Small Cap US Stocks / S&P SmallCap 600 -- IJR (iShares S&P SmallCap 600 ETF)
Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)
Small Cap US Stocks / S&P SmallCap 600 -- IJR (iShares S&P SmallCap 600 ETF)
Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)
Small Cap US Stocks / S&P SmallCap 600 -- IJR (iShares S&P SmallCap 600 ETF)
Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)
US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)
US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)
US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)
US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)
US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)
Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)
Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)

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[illegible]

Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)
 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
 Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)
 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
 Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)
 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
 Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)
 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
 Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)
 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
 Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)
 Gold -- GLD (SPDR Gold Shares)
 Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)
 Gold -- GLD (SPDR Gold Shares)
 Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)
 Gold -- GLD (SPDR Gold Shares)
 Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)
 Gold -- GLD (SPDR Gold Shares)
 Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)
 Gold -- GLD (SPDR Gold Shares)
 Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
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 Error creating temp_df for IVV and ETH-USD: "['ETH-USD_Daily_Return'] not in index"
 Mid Cap US Stocks / S&P MidCap 400 -- IJH (iShares S&P MidCap 400 ETF)
 Small Cap US Stocks / S&P SmallCap 600 -- IJR (iShares S&P SmallCap 600 ETF)
 Mid Cap US Stocks / S&P MidCap 400 -- IJH (iShares S&P MidCap 400 ETF)
 Small Cap US Stocks / S&P SmallCap 600 -- IJR (iShares S&P SmallCap 600 ETF)

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[illegible]

[illegible]

Mid Cap US Stocks / S&P MidCap 400 -- IJH (iShares S&P MidCap 400 ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Mid Cap US Stocks / S&P MidCap 400 -- IJH (iShares S&P MidCap 400 ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Mid Cap US Stocks / S&P MidCap 400 -- IJH (iShares S&P MidCap 400 ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Mid Cap US Stocks / S&P MidCap 400 -- IJH (iShares S&P MidCap 400 ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Mid Cap US Stocks / S&P MidCap 400 -- IJH (iShares S&P MidCap 400 ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Mid Cap US Stocks / S&P MidCap 400 -- IJH (iShares S&P MidCap 400 ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Mid Cap US Stocks / S&P MidCap 400 -- IJH (iShares S&P MidCap 400 ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Error creating temp_df for IJH and BTC-USD: "['BTC-USD_Daily_Return'] not in index"
 Error creating temp_df for IJH and ETH-USD: "['ETH-USD_Daily_Return'] not in index"
 Small Cap US Stocks / S&P SmallCap 600 -- IJR (iShares S&P SmallCap 600 ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Small Cap US Stocks / S&P SmallCap 600 -- IJR (iShares S&P SmallCap 600 ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Small Cap US Stocks / S&P SmallCap 600 -- IJR (iShares S&P SmallCap 600 ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Small Cap US Stocks / S&P SmallCap 600 -- IJR (iShares S&P SmallCap 600 ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Small Cap US Stocks / S&P SmallCap 600 -- IJR (iShares S&P SmallCap 600 ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Small Cap US Stocks / S&P SmallCap 600 -- IJR (iShares S&P SmallCap 600 ETF)
 Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
 Small Cap US Stocks / S&P SmallCap 600 -- IJR (iShares S&P SmallCap 600 ETF)
 Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
 Small Cap US Stocks / S&P SmallCap 600 -- IJR (iShares S&P SmallCap 600 ETF)
 Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
 Small Cap US Stocks / S&P SmallCap 600 -- IJR (iShares S&P SmallCap 600 ETF)
 Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
 Small Cap US Stocks / S&P SmallCap 600 -- IJR (iShares S&P SmallCap 600 ETF)
 Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
 Small Cap US Stocks / S&P SmallCap 600 -- IJR (iShares S&P SmallCap 600 ETF)
 Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Small Cap US Stocks / S&P SmallCap 600 -- IJR (iShares S&P SmallCap 600 ETF)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Small Cap US Stocks / S&P SmallCap 600 -- IJR (iShares S&P SmallCap 600 ETF)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Small Cap US Stocks / S&P SmallCap 600 -- IJR (iShares S&P SmallCap 600 ETF)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
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 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Small Cap US Stocks / S&P SmallCap 600 -- IJR (iShares S&P SmallCap 600 ETF)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)

Small Cap US Stocks / S&P SmallCap 600 -- IJR (iShares S&P SmallCap 600 ETF)
 Gold -- GLD (SPDR Gold Shares)
 Small Cap US Stocks / S&P SmallCap 600 -- IJR (iShares S&P SmallCap 600 ETF)
 Gold -- GLD (SPDR Gold Shares)
 Small Cap US Stocks / S&P SmallCap 600 -- IJR (iShares S&P SmallCap 600 ETF)
 Gold -- GLD (SPDR Gold Shares)
 Small Cap US Stocks / S&P SmallCap 600 -- IJR (iShares S&P SmallCap 600 ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Small Cap US Stocks / S&P SmallCap 600 -- IJR (iShares S&P SmallCap 600 ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Small Cap US Stocks / S&P SmallCap 600 -- IJR (iShares S&P SmallCap 600 ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Small Cap US Stocks / S&P SmallCap 600 -- IJR (iShares S&P SmallCap 600 ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Small Cap US Stocks / S&P SmallCap 600 -- IJR (iShares S&P SmallCap 600 ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Small Cap US Stocks / S&P SmallCap 600 -- IJR (iShares S&P SmallCap 600 ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Small Cap US Stocks / S&P SmallCap 600 -- IJR (iShares S&P SmallCap 600 ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Small Cap US Stocks / S&P SmallCap 600 -- IJR (iShares S&P SmallCap 600 ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Small Cap US Stocks / S&P SmallCap 600 -- IJR (iShares S&P SmallCap 600 ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Small Cap US Stocks / S&P SmallCap 600 -- IJR (iShares S&P SmallCap 600 ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Error creating temp_df for IJR and BTC-USD: "['BTC-USD_Daily_Return'] not in index"
 Error creating temp_df for IJR and ETH-USD: "['ETH-USD_Daily_Return'] not in index"
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)

US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)

 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 International Developed Market Stocks -- EFA (iShares MSCI EAFE ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 International Developed Market Stocks -- EFA (iShares MSCI EAFE ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 International Developed Market Stocks -- EFA (iShares MSCI EAFE ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 International Developed Market Stocks -- EFA (iShares MSCI EAFE ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 International Developed Market Stocks -- EFA (iShares MSCI EAFE ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 European Stocks -- IEV (iShares S&P Europe 350 ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 European Stocks -- IEV (iShares S&P Europe 350 ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)

European Stocks -- IEV (iShares S&P Europe 350 ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 European Stocks -- IEV (iShares S&P Europe 350 ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 European Stocks -- IEV (iShares S&P Europe 350 ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Gold -- GLD (SPDR Gold Shares)

US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Gold -- GLD (SPDR Gold Shares)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Gold -- GLD (SPDR Gold Shares)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Gold -- GLD (SPDR Gold Shares)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Gold -- GLD (SPDR Gold Shares)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
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 Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)
 Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)
 Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)


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Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)

Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
Gold -- GLD (SPDR Gold Shares)
Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
Gold -- GLD (SPDR Gold Shares)
Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
Gold -- GLD (SPDR Gold Shares)
Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
Gold -- GLD (SPDR Gold Shares)
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Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)

 Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
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 Error creating temp_df for IWB and ETH-USD: "['ETH-USD_Daily_Return'] not in index"
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 International Developed Market Stocks -- EFA (iShares MSCI EAFE ETF)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 International Developed Market Stocks -- EFA (iShares MSCI EAFE ETF)

Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Gold -- GLD (SPDR Gold Shares)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Gold -- GLD (SPDR Gold Shares)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Gold -- GLD (SPDR Gold Shares)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Gold -- GLD (SPDR Gold Shares)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Gold -- GLD (SPDR Gold Shares)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Gold -- GLD (SPDR Gold Shares)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)

Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Error creating temp_df for IWM and BTC-USD: "['BTC-USD_Daily_Return'] not in index"
 Error creating temp_df for IWM and ETH-USD: "['ETH-USD_Daily_Return'] not in index"
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)
 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)
 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)
 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)
 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)
 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)
 International Developed Market Stocks -- EFA (iShares MSCI EAFE ETF)
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)
 International Developed Market Stocks -- EFA (iShares MSCI EAFE ETF)
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)
 International Developed Market Stocks -- EFA (iShares MSCI EAFE ETF)
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)
 International Developed Market Stocks -- EFA (iShares MSCI EAFE ETF)
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)
 International Developed Market Stocks -- EFA (iShares MSCI EAFE ETF)
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)
 Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)
 Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)
 Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)
 Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)
 Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)
 European Stocks -- IEV (iShares S&P Europe 350 ETF)
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)
 European Stocks -- IEV (iShares S&P Europe 350 ETF)
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)
 European Stocks -- IEV (iShares S&P Europe 350 ETF)
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)
 European Stocks -- IEV (iShares S&P Europe 350 ETF)
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)
 European Stocks -- IEV (iShares S&P Europe 350 ETF)
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)
 European Stocks -- IEV (iShares S&P Europe 350 ETF)
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)

Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)
 Gold -- GLD (SPDR Gold Shares)
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Error creating temp_df for IWD and BTC-USD: "['BTC-USD_Daily_Return'] not in index"
 Error creating temp_df for IWD and ETH-USD: "['ETH-USD_Daily_Return'] not in index"
 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)
 International Developed Market Stocks -- EFA (iShares MSCI EAFE ETF)
 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)
 International Developed Market Stocks -- EFA (iShares MSCI EAFE ETF)
 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)
 International Developed Market Stocks -- EFA (iShares MSCI EAFE ETF)
 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)
 International Developed Market Stocks -- EFA (iShares MSCI EAFE ETF)
 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)
 International Developed Market Stocks -- EFA (iShares MSCI EAFE ETF)
 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)
 International Developed Market Stocks -- EFA (iShares MSCI EAFE ETF)
 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)
 Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)
 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)
 Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)
 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)
 Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)
 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)
 Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)
 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)
 Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)
 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)
 Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)
 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)

Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)
 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)
 Gold -- GLD (SPDR Gold Shares)
 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)
 Gold -- GLD (SPDR Gold Shares)
 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)
 Gold -- GLD (SPDR Gold Shares)
 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)
 Gold -- GLD (SPDR Gold Shares)
 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)
 Gold -- GLD (SPDR Gold Shares)
 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)

 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Error creating temp_df for IWF and BTC-USD: "['BTC-USD_Daily_Return'] not in index"
 Error creating temp_df for IWF and ETH-USD: "['ETH-USD_Daily_Return'] not in index"
 International Developed Market Stocks -- EFA (iShares MSCI EAFE ETF)
 Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)
 International Developed Market Stocks -- EFA (iShares MSCI EAFE ETF)
 Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)
 International Developed Market Stocks -- EFA (iShares MSCI EAFE ETF)
 Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)
 International Developed Market Stocks -- EFA (iShares MSCI EAFE ETF)
 Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)
 International Developed Market Stocks -- EFA (iShares MSCI EAFE ETF)
 Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)
 International Developed Market Stocks -- EFA (iShares MSCI EAFE ETF)
 Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)
 International Developed Market Stocks -- EFA (iShares MSCI EAFE ETF)

International Developed Market Stocks -- EFA (iShares MSCI EAFE ETF)
 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
 International Developed Market Stocks -- EFA (iShares MSCI EAFE ETF)
 Gold -- GLD (SPDR Gold Shares)
 International Developed Market Stocks -- EFA (iShares MSCI EAFE ETF)
 Gold -- GLD (SPDR Gold Shares)
 International Developed Market Stocks -- EFA (iShares MSCI EAFE ETF)
 Gold -- GLD (SPDR Gold Shares)
 International Developed Market Stocks -- EFA (iShares MSCI EAFE ETF)
 Gold -- GLD (SPDR Gold Shares)
 International Developed Market Stocks -- EFA (iShares MSCI EAFE ETF)
 Gold -- GLD (SPDR Gold Shares)
 International Developed Market Stocks -- EFA (iShares MSCI EAFE ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 International Developed Market Stocks -- EFA (iShares MSCI EAFE ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 International Developed Market Stocks -- EFA (iShares MSCI EAFE ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 International Developed Market Stocks -- EFA (iShares MSCI EAFE ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)

 International Developed Market Stocks -- EFA (iShares MSCI EAFE ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 International Developed Market Stocks -- EFA (iShares MSCI EAFE ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 International Developed Market Stocks -- EFA (iShares MSCI EAFE ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 International Developed Market Stocks -- EFA (iShares MSCI EAFE ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 International Developed Market Stocks -- EFA (iShares MSCI EAFE ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 International Developed Market Stocks -- EFA (iShares MSCI EAFE ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
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 Error creating temp_df for EFA and ETH-USD: "['ETH-USD_Daily_Return'] not in index"
 Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)
 European Stocks -- IEV (iShares S&P Europe 350 ETF)
 Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)
 European Stocks -- IEV (iShares S&P Europe 350 ETF)
 Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)
 European Stocks -- IEV (iShares S&P Europe 350 ETF)
 Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)
 European Stocks -- IEV (iShares S&P Europe 350 ETF)
 Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)
 European Stocks -- IEV (iShares S&P Europe 350 ETF)
 Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)

[illegible]

Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)
 Gold -- GLD (SPDR Gold Shares)
 Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Error creating temp_df for EEM and BTC-USD: "['BTC-USD_Daily_Return'] not in index"
 Error creating temp_df for EEM and ETH-USD: "['ETH-USD_Daily_Return'] not in index"
 European Stocks -- IEV (iShares S&P Europe 350 ETF)
 Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
 European Stocks -- IEV (iShares S&P Europe 350 ETF)
 Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
 European Stocks -- IEV (iShares S&P Europe 350 ETF)
 Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
 European Stocks -- IEV (iShares S&P Europe 350 ETF)
 Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
 European Stocks -- IEV (iShares S&P Europe 350 ETF)
 Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
 European Stocks -- IEV (iShares S&P Europe 350 ETF)
 Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
 European Stocks -- IEV (iShares S&P Europe 350 ETF)
 Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
 European Stocks -- IEV (iShares S&P Europe 350 ETF)
 Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
 European Stocks -- IEV (iShares S&P Europe 350 ETF)
 Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
 European Stocks -- IEV (iShares S&P Europe 350 ETF)
 Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
 European Stocks -- IEV (iShares S&P Europe 350 ETF)
 Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
 European Stocks -- IEV (iShares S&P Europe 350 ETF)
 Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
 European Stocks -- IEV (iShares S&P Europe 350 ETF)

European Stocks -- IEV (iShares S&P Europe 350 ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Error creating temp_df for IEV and BTC-USD: "['BTC-USD_Daily_Return'] not in index"
 Error creating temp_df for IEV and ETH-USD: "['ETH-USD_Daily_Return'] not in index"
 Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
 Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
 Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
 Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
 Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
 Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
 Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
 Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
 Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
 Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
 Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
 Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
 Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
 Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
 Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
 Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
 Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
 Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
 Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
 Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
 Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
 Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
 Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
 Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
 Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
 Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
 Gold -- GLD (SPDR Gold Shares)
 Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
 Gold -- GLD (SPDR Gold Shares)
 Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
 Gold -- GLD (SPDR Gold Shares)
 Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
 Gold -- GLD (SPDR Gold Shares)
 Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
 Gold -- GLD (SPDR Gold Shares)
 Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
 Gold -- GLD (SPDR Gold Shares)

Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Error creating temp_df for SHY and BTC-USD: "['BTC-USD_Daily_Return'] not in index"
 Error creating temp_df for SHY and ETH-USD: "['ETH-USD_Daily_Return'] not in index"
 Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
 Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
 Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
 Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
 Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
 Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
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 Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
 Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
 Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
 Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
 Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
 Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
 Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
 Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
 Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
 Gold -- GLD (SPDR Gold Shares)
 Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
 Gold -- GLD (SPDR Gold Shares)

Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
 Gold -- GLD (SPDR Gold Shares)
 Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
 Gold -- GLD (SPDR Gold Shares)
 Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
 Gold -- GLD (SPDR Gold Shares)
 Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
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 Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)

 Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Error creating temp_df for IEF and BTC-USD: "['BTC-USD_Daily_Return'] not in index"
 Error creating temp_df for IEF and ETH-USD: "['ETH-USD_Daily_Return'] not in index"
 Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
 Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
 Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
 Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
 Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
 Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
 Gold -- GLD (SPDR Gold Shares)
 Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
 Gold -- GLD (SPDR Gold Shares)
 Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
 Gold -- GLD (SPDR Gold Shares)
 Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
 Gold -- GLD (SPDR Gold Shares)

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 Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
 Gold -- GLD (SPDR Gold Shares)
 Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)

 Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
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 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF)
 Real Estate -- IYR (iShares U.S. Real Estate ETF)
 Error creating temp_df for TLT and BTC-USD: "['BTC-USD_Daily_Return'] not in index"
 Error creating temp_df for TLT and ETH-USD: "['ETH-USD_Daily_Return'] not in index"
 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
 Gold -- GLD (SPDR Gold Shares)
 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
 Gold -- GLD (SPDR Gold Shares)
 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
 Gold -- GLD (SPDR Gold Shares)
 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
 Gold -- GLD (SPDR Gold Shares)
 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
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 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
 Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
 Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)

```

Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
Real Estate -- IYR (iShares U.S. Real Estate ETF)
Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
Real Estate -- IYR (iShares U.S. Real Estate ETF)
Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
Real Estate -- IYR (iShares U.S. Real Estate ETF)
Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
Real Estate -- IYR (iShares U.S. Real Estate ETF)
Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)
Real Estate -- IYR (iShares U.S. Real Estate ETF)
Error creating temp_df for AGG and BTC-USD: '['BTC-USD_Daily_Return'] not in
index"
Error creating temp_df for AGG and ETH-USD: '['ETH-USD_Daily_Return'] not in
index"
Gold -- GLD (SPDR Gold Shares)
Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
Gold -- GLD (SPDR Gold Shares)
Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
Gold -- GLD (SPDR Gold Shares)
Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
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Gold -- GLD (SPDR Gold Shares)
Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
Gold -- GLD (SPDR Gold Shares)
Real Estate -- IYR (iShares U.S. Real Estate ETF)
Gold -- GLD (SPDR Gold Shares)
Real Estate -- IYR (iShares U.S. Real Estate ETF)
Gold -- GLD (SPDR Gold Shares)
Real Estate -- IYR (iShares U.S. Real Estate ETF)
Gold -- GLD (SPDR Gold Shares)
Real Estate -- IYR (iShares U.S. Real Estate ETF)
Gold -- GLD (SPDR Gold Shares)
Real Estate -- IYR (iShares U.S. Real Estate ETF)
Gold -- GLD (SPDR Gold Shares)
Real Estate -- IYR (iShares U.S. Real Estate ETF)
Error creating temp_df for GLD and BTC-USD: '['BTC-USD_Daily_Return'] not in
index"
Error creating temp_df for GLD and ETH-USD: '['ETH-USD_Daily_Return'] not in
index"
Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
Real Estate -- IYR (iShares U.S. Real Estate ETF)
Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
Real Estate -- IYR (iShares U.S. Real Estate ETF)

Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
Real Estate -- IYR (iShares U.S. Real Estate ETF)
Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)
Real Estate -- IYR (iShares U.S. Real Estate ETF)
Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)

```

```

Real Estate -- IYR (iShares U.S. Real Estate ETF)
Error creating temp_df for GSG and BTC-USD: "['BTC-USD_Daily_Return'] not in
index"
Error creating temp_df for GSG and ETH-USD: "['ETH-USD_Daily_Return'] not in
index"
Error creating temp_df for IYR and BTC-USD: "['BTC-USD_Daily_Return'] not in
index"
Error creating temp_df for IYR and ETH-USD: "['ETH-USD_Daily_Return'] not in
index"
Error creating temp_df for BTC-USD and ETH-USD: "None of [Index(['BTC-
USD_Daily_Return', 'ETH-USD_Daily_Return'], dtype='object')] are in the
[columns]"

```

```
[16]: display(rolling_correlation_results_no_crypto_df)
```

	IVV_IJH_1mo	IVV_IJH_3mo	IVV_IJH_6mo	IVV_IJH_1y	IVV_IJH_5y	\
Date						
2006-07-24	NaN	NaN	NaN	NaN	NaN	
2006-07-25	NaN	NaN	NaN	NaN	NaN	
2006-07-26	NaN	NaN	NaN	NaN	NaN	
2006-07-27	NaN	NaN	NaN	NaN	NaN	
2006-07-28	NaN	NaN	NaN	NaN	NaN	
...	...	...	...	...	...	
2026-04-24	0.89	0.85	0.82	0.82	0.88	
2026-04-27	0.88	0.85	0.82	0.82	0.88	
2026-04-28	0.86	0.85	0.82	0.82	0.88	
2026-04-29	0.85	0.85	0.82	0.82	0.88	
2026-04-30	0.78	0.85	0.83	0.82	0.88	

	IVV_IJR_1mo	IVV_IJR_3mo	IVV_IJR_6mo	IVV_IJR_1y	IVV_IJR_5y	\
Date						
2006-07-24	NaN	NaN	NaN	NaN	NaN	
2006-07-25	NaN	NaN	NaN	NaN	NaN	
2006-07-26	NaN	NaN	NaN	NaN	NaN	
2006-07-27	NaN	NaN	NaN	NaN	NaN	
2006-07-28	NaN	NaN	NaN	NaN	NaN	
...	...	...	...	...	...	
2026-04-24	0.92	0.85	0.79	0.79	0.82	
2026-04-27	0.91	0.85	0.79	0.79	0.82	
2026-04-28	0.90	0.86	0.79	0.79	0.82	
2026-04-29	0.89	0.86	0.80	0.79	0.82	
2026-04-30	0.83	0.86	0.80	0.79	0.82	

	GLD_IYR_1mo	GLD_IYR_3mo	GLD_IYR_6mo	GLD_IYR_1y	\
Date					
2006-07-24	NaN	NaN	NaN	NaN	
2006-07-25	NaN	NaN	NaN	NaN	
2006-07-26	NaN	NaN	NaN	NaN	

2006-07-27	...	NaN	NaN	NaN	NaN
2006-07-28	...	NaN	NaN	NaN	NaN
...	...	...	...	...	...
2026-04-24	...	0.25	0.22	0.14	0.11
2026-04-27	...	0.24	0.22	0.15	0.11
2026-04-28	...	0.38	0.22	0.14	0.11
2026-04-29	...	0.41	0.26	0.15	0.11
2026-04-30	...	0.36	0.27	0.15	0.11

	GLD_IYR_5y	GSG_IYR_1mo	GSG_IYR_3mo	GSG_IYR_6mo	GSG_IYR_1y	\
Date						
2006-07-24	NaN	NaN	NaN	NaN	NaN	
2006-07-25	NaN	NaN	NaN	NaN	NaN	
2006-07-26	NaN	NaN	NaN	NaN	NaN	
2006-07-27	NaN	NaN	NaN	NaN	NaN	
2006-07-28	NaN	NaN	NaN	NaN	NaN	
...	...	...	...	...	...	
2026-04-24	0.18	-0.45	-0.35	-0.17	-0.17	
2026-04-27	0.18	-0.44	-0.35	-0.17	-0.17	
2026-04-28	0.18	-0.37	-0.34	-0.17	-0.17	
2026-04-29	0.18	-0.43	-0.35	-0.18	-0.17	
2026-04-30	0.18	-0.42	-0.37	-0.21	-0.17	

	GSG_IYR_5y
Date	
2006-07-24	NaN
2006-07-25	NaN
2006-07-26	NaN
2006-07-27	NaN
2006-07-28	NaN
...	...
2026-04-24	0.11
2026-04-27	0.11
2026-04-28	0.11
2026-04-29	0.11
2026-04-30	0.10

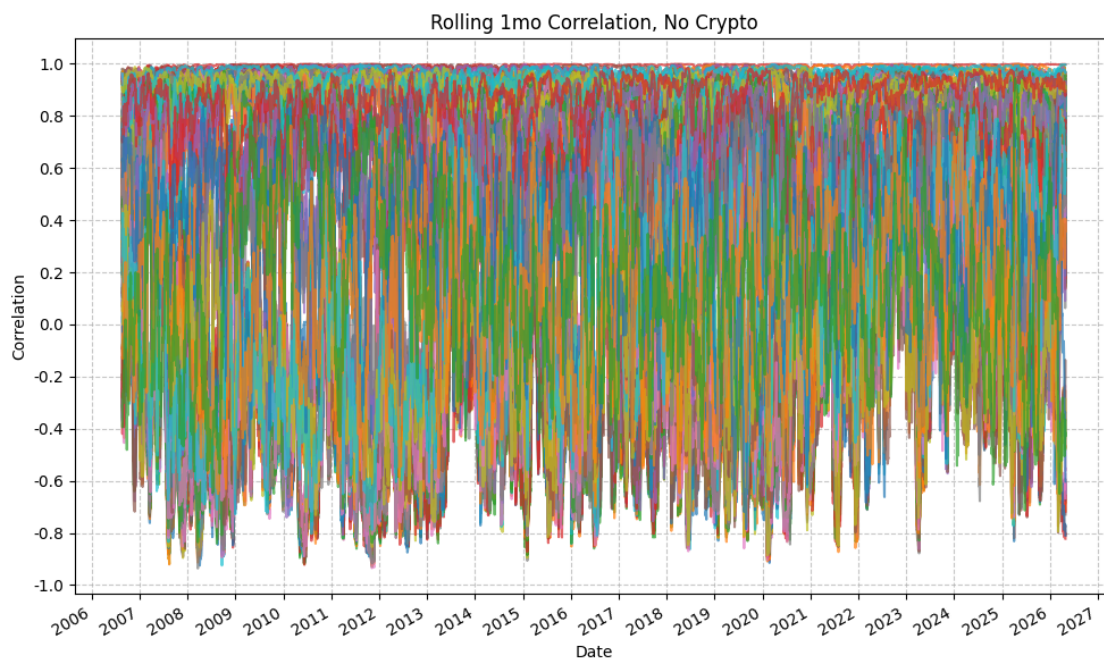
[4974 rows x 765 columns]

```
[17]: plot_time_series(
    df=rolling_correlation_results_no_crypto_df,
    plot_start_date=None,
    plot_end_date=None,
    plot_columns=[col for col in rolling_correlation_results_no_crypto_df.
↪columns if "1mo" in col],
    title="Rolling 1mo Correlation, No Crypto",
    x_label="Date",
```

```

x_format="Year",
x_tick_spacing=1,
x_tick_rotation=30,
y_label="Correlation",
y_format="Decimal",
y_format_decimal_places="Auto",
y_tick_spacing="Auto",
y_tick_rotation=0,
grid=True,
legend=False,
export_plot=False,
plot_file_name=None,
)

```



```

[18]: # plot_time_series(
#     df=rolling_correlation_results_no_crypto_df,
#     plot_start_date=None,
#     plot_end_date=None,
#     plot_columns=[
#         col for col in rolling_correlation_results_no_crypto_df.columns
#         if "1y" in col
#         and any(col.startswith(ticker) for ticker in equity_tickers)
#         and not any(col.endswith(f"{ticker}_1y") for ticker in bond_tickers +
#             ↪commodity_tickers + real_estate_tickers)
#     ],

```

```
# title="Rolling 1y Correlation - Equity ETFs",
# x_label="Date",
# x_format="Year",
# x_tick_spacing=1,
# x_tick_rotation=30,
# y_label="Correlation",
# y_format="Decimal",
# y_format_decimal_places="Auto",
# y_tick_spacing="Auto",
# y_tick_rotation=0,
# grid=True,
# legend=True,
# export_plot=False,
# plot_file_name=None,
# )
```

```
[19]: # plot_time_series(
# df=rolling_correlation_results_no_crypto_df,
# plot_start_date=None,
# plot_end_date=None,
# plot_columns=[
#     col for col in rolling_correlation_results_no_crypto_df.columns
#     if "1y" in col
#     and any(col.startswith(ticker) for ticker in us_equity_tickers)
#     and not any(col.endswith(f"{ticker}_1y") for ticker in
↳ intl_equity_tickers + bond_tickers + commodity_tickers + real_estate_tickers)
# ],
# title="Rolling 1y Correlation - US Equity ETFs",
# x_label="Date",
# x_format="Year",
# x_tick_spacing=1,
# x_tick_rotation=30,
# y_label="Correlation",
# y_format="Decimal",
# y_format_decimal_places="Auto",
# y_tick_spacing="Auto",
# y_tick_rotation=0,
# grid=True,
# legend=True,
# export_plot=False,
# plot_file_name=None,
# )
```

```
[20]: # plot_time_series(
# df=rolling_correlation_results_no_crypto_df,
# plot_start_date=None,
# plot_end_date=None,
```

```

#     plot_columns=[
#         col for col in rolling_correlation_results_no_crypto_df.columns
#         if "6mo" in col
#         and any(col.startswith(ticker) for ticker in us_equity_tickers)
#         and not any(col.startswith("QQQ") for ticker in us_equity_tickers)
#         and not any(col.endswith(f"{ticker}_6mo") for ticker in
↳ intl_equity_tickers + bond_tickers + commodity_tickers + real_estate_tickers
↳ + list("QQQ" + "IWD" + "IWF"))
#     ],
#     title="Rolling 6mo Correlation - US Equity ETFs",
#     x_label="Date",
#     x_format="Year",
#     x_tick_spacing=1,
#     x_tick_rotation=30,
#     y_label="Correlation",
#     y_format="Decimal",
#     y_format_decimal_places="Auto",
#     y_tick_spacing="Auto",
#     y_tick_rotation=0,
#     grid=True,
#     legend=True,
#     export_plot=False,
#     plot_file_name=None,
# )

```

```

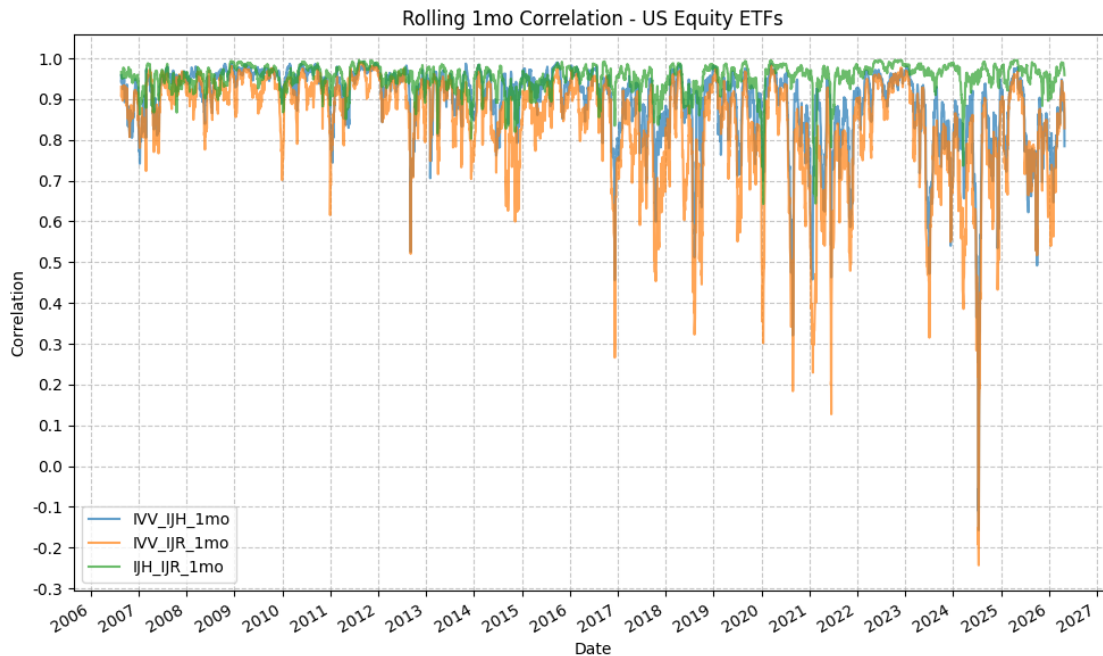
[21]: plot_time_series(
    df=rolling_correlation_results_no_crypto_df,
    plot_start_date=None,
    plot_end_date=None,
    plot_columns=[
        col for col in rolling_correlation_results_no_crypto_df.columns
        if "1mo" in col
        and any(col.startswith(ticker) for ticker in us_equity_tickers)
        and not any(col.startswith("QQQ") for ticker in us_equity_tickers)
        and not any(col.endswith(f"{ticker}_1mo") for ticker in
↳ intl_equity_tickers + bond_tickers + commodity_tickers + real_estate_tickers
↳ + list("QQQ" + "IWD" + "IWF" + "IWB" + "IWM"))
    ],
    title="Rolling 1mo Correlation - US Equity ETFs",
    x_label="Date",
    x_format="Year",
    x_tick_spacing=1,
    x_tick_rotation=30,
    y_label="Correlation",
    y_format="Decimal",
    y_format_decimal_places="Auto",
    y_tick_spacing="Auto",

```

```

y_tick_rotation=0,
grid=True,
legend=True,
export_plot=False,
plot_file_name=None,
)

```



0.6 Rolling Correlations Amongst US Stocks

We'll now look specifically at the rolling correlations between the US S&P index ETFs (IVV, IJH, and IJR). We'll isolate the data for the rolling correlations for these three assets and see how the correlations have changed over time.

```

[22]: corr_list = ['IVV_IJH_1mo', 'IVV_IJR_1mo', 'IJH_IJR_1mo']

plot_time_series(
    df=rolling_correlation_results_no_crypto_df,
    plot_start_date=None,
    plot_end_date=None,
    plot_columns=corr_list,
    title="Rolling 6mo Correlation - US Equity ETFs (S&P 500, 400, 600)",
    x_label="Date",
    x_format="Year",
    x_tick_spacing=1,
    x_tick_rotation=30,
)

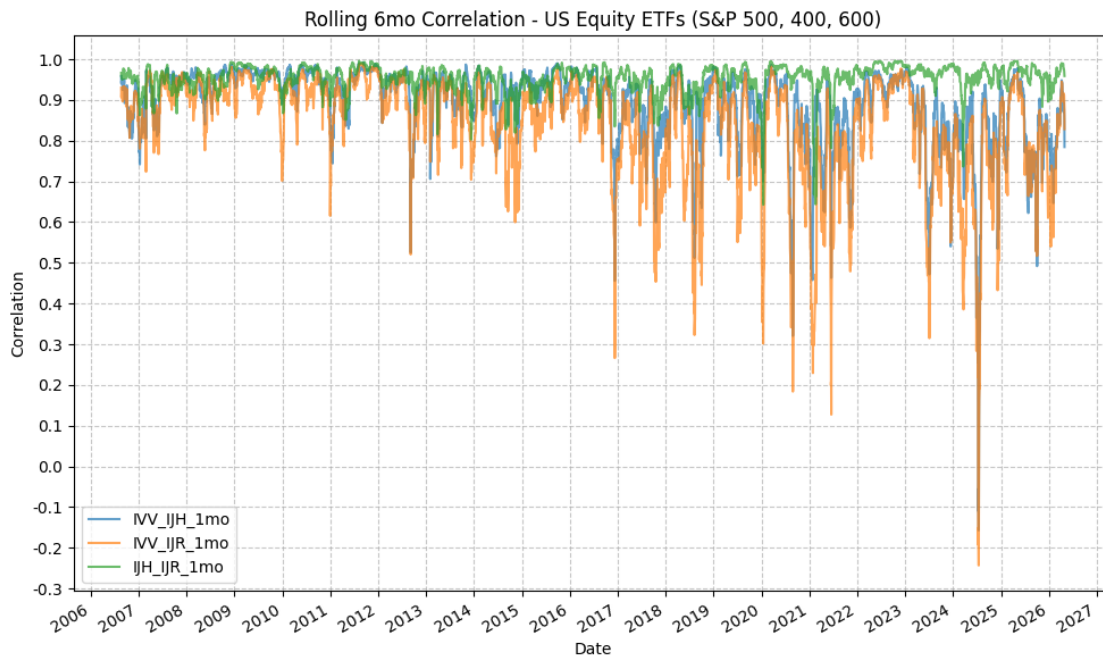
```



```

y_label="Correlation",
y_format="Decimal",
y_format_decimal_places="Auto",
y_tick_spacing="Auto",
y_tick_rotation=0,
grid=True,
legend=True,
export_plot=False,
plot_file_name=None,
)

```



With the above plot, we can see that there are periods of time when the correlations abruptly increase, such as:

- Early-mid 2007 (the start of the financial crisis)
- Mid 2011 (the European debt crisis)
- Late 2015 into 2016
- Late 2018 (rate hikes?)
- Early 2020 (COVID-19 pandemic)
- Late 2022 into early 2023 (rate hikes, recession fears, COVID tech bubble, etc.)
- Mid 2024 (banking crisis, rate hikes)
- Early 2025 (Liberation day)

We're not necessarily looking to explain away each of these time periods or delve into the macro factors that may have been at play or driving the correlations, but we are more interested in the change in correlation over time on response to some kind of market shock or event.

As an attempt to find some kind of signal, let's simply add the three correlations together to get a "total correlation" metric, and then plot that total correlation metric over time to see if we can identify any trends or patterns.

```
[23]: us_sp_etfs = rolling_correlation_results_no_crypto_df[corr_list].dropna()

# Add the correlations together
us_sp_etfs["total_correlation"] = us_sp_etfs.sum(axis=1)

# Calc the percent change in total correlation
us_sp_etfs["total_correlation_pct_change"] = us_sp_etfs["total_correlation"].
    pct_change()

# Merge daily returns into us_sp_etfs for the three ETFs
us_sp_etfs = us_sp_etfs.merge(
    fund_data_daily_returns[["IVV_Daily_Return", "IJH_Daily_Return",
        "IJR_Daily_Return"]],
    left_index=True,
    right_index=True,
    how="left"
)

windows = [5, 10, 15, 20, 25]
for window in windows:
    us_sp_etfs[f"total_correlation_pct_change_{window}d_cum"] =
    us_sp_etfs["total_correlation_pct_change"].rolling(window=window).
    apply(lambda x: (1 + x).prod() - 1)
    us_sp_etfs[f"IVV_Daily_Return_{window}d_cum"] =
    us_sp_etfs["IVV_Daily_Return"].rolling(window=window).apply(lambda x: (1 +
    x).prod() - 1)
    us_sp_etfs[f"IJH_Daily_Return_{window}d_cum"] =
    us_sp_etfs["IJH_Daily_Return"].rolling(window=window).apply(lambda x: (1 +
    x).prod() - 1)
    us_sp_etfs[f"IJR_Daily_Return_{window}d_cum"] =
    us_sp_etfs["IJR_Daily_Return"].rolling(window=window).apply(lambda x: (1 +
    x).prod() - 1)

pandas.set_decimal_places(4)
display(us_sp_etfs)
```

	IVV_IJH_1mo	IVV_IJR_1mo	IJH_IJR_1mo	total_correlation \
Date				
2006-08-21	0.9585	0.9301	0.9665	2.8552
2006-08-22	0.9403	0.9066	0.9564	2.8034
2006-08-23	0.9462	0.9109	0.9574	2.8144
2006-08-24	0.9442	0.9206	0.9623	2.8271
2006-08-25	0.9515	0.9257	0.9578	2.8350
...	...	...	...	...

2026-04-24	0.8906	0.9159	0.9755	2.7820
2026-04-27	0.8768	0.9120	0.9730	2.7618
2026-04-28	0.8601	0.8967	0.9685	2.7253
2026-04-29	0.8529	0.8854	0.9677	2.7060
2026-04-30	0.7845	0.8284	0.9588	2.5717

	total_correlation_pct_change	IVV_Daily_Return	IJH_Daily_Return	\
Date				
2006-08-21	NaN	NaN	NaN	
2006-08-22	-0.0181	NaN	NaN	
2006-08-23	0.0040	NaN	NaN	
2006-08-24	0.0045	NaN	NaN	
2006-08-25	0.0028	NaN	NaN	
...	...	...	...	
2026-04-24	-0.0001	0.0078	0.0021	
2026-04-27	-0.0073	0.0017	0.0007	
2026-04-28	-0.0132	-0.0049	-0.0100	
2026-04-29	-0.0071	-0.0001	-0.0075	
2026-04-30	-0.0496	0.0100	0.0168	

	IJR_Daily_Return	total_correlation_pct_change_5d_cum	\
Date			
2006-08-21	NaN	NaN	
2006-08-22	NaN	NaN	
2006-08-23	NaN	NaN	
2006-08-24	NaN	NaN	
2006-08-25	NaN	NaN	
...	...	...	
2026-04-24	0.0054	-0.0104	
2026-04-27	0.0019	-0.0156	
2026-04-28	-0.0056	-0.0223	
2026-04-29	-0.0078	-0.0132	
2026-04-30	0.0176	-0.0757	

	IVV_Daily_Return_5d_cum	...	IJH_Daily_Return_15d_cum	\
Date				
2006-08-21	NaN	...	NaN	
2006-08-22	NaN	...	NaN	
2006-08-23	NaN	...	NaN	
2006-08-24	NaN	...	NaN	
2006-08-25	NaN	...	NaN	
...	...	...	...	
2026-04-24	0.0055	...	0.0676	
2026-04-27	0.0090	...	0.0640	
2026-04-28	0.0109	...	0.0519	
2026-04-29	0.0005	...	0.0147	
2026-04-30	0.0145	...	0.0297	

	IJR_Daily_Return_15d_cum	total_correlation_pct_change_20d_cum \
Date		
2006-08-21	NaN	NaN
2006-08-22	NaN	NaN
2006-08-23	NaN	NaN
2006-08-24	NaN	NaN
2006-08-25	NaN	NaN
...	...	...
2026-04-24	0.0867	0.0496
2026-04-27	0.0829	0.0223
2026-04-28	0.0733	0.0049
2026-04-29	0.0377	-0.0318
2026-04-30	0.0486	-0.0816

	IVV_Daily_Return_20d_cum	IJH_Daily_Return_20d_cum \
Date		
2006-08-21	NaN	NaN
2006-08-22	NaN	NaN
2006-08-23	NaN	NaN
2006-08-24	NaN	NaN
2006-08-25	NaN	NaN
...	...	...
2026-04-24	0.1069	0.0830
2026-04-27	0.1280	0.1018
2026-04-28	0.1260	0.0994
2026-04-29	0.0944	0.0598
2026-04-30	0.0973	0.0686

	IJR_Daily_Return_20d_cum	total_correlation_pct_change_25d_cum \
Date		
2006-08-21	NaN	NaN
2006-08-22	NaN	NaN
2006-08-23	NaN	NaN
2006-08-24	NaN	NaN
2006-08-25	NaN	NaN
...	...	...
2026-04-24	0.1021	0.0592
2026-04-27	0.1216	0.0343
2026-04-28	0.1234	0.0089
2026-04-29	0.0838	0.0224
2026-04-30	0.0975	-0.0274

	IVV_Daily_Return_25d_cum	IJH_Daily_Return_25d_cum \
Date		
2006-08-21	NaN	NaN
2006-08-22	NaN	NaN
2006-08-23	NaN	NaN
2006-08-24	NaN	NaN

2006-08-25	NaN	NaN
...	...	...
2026-04-24	0.0853	0.0808
2026-04-27	0.1032	0.1058
2026-04-28	0.0860	0.0750
2026-04-29	0.0894	0.0583
2026-04-30	0.0945	0.0667

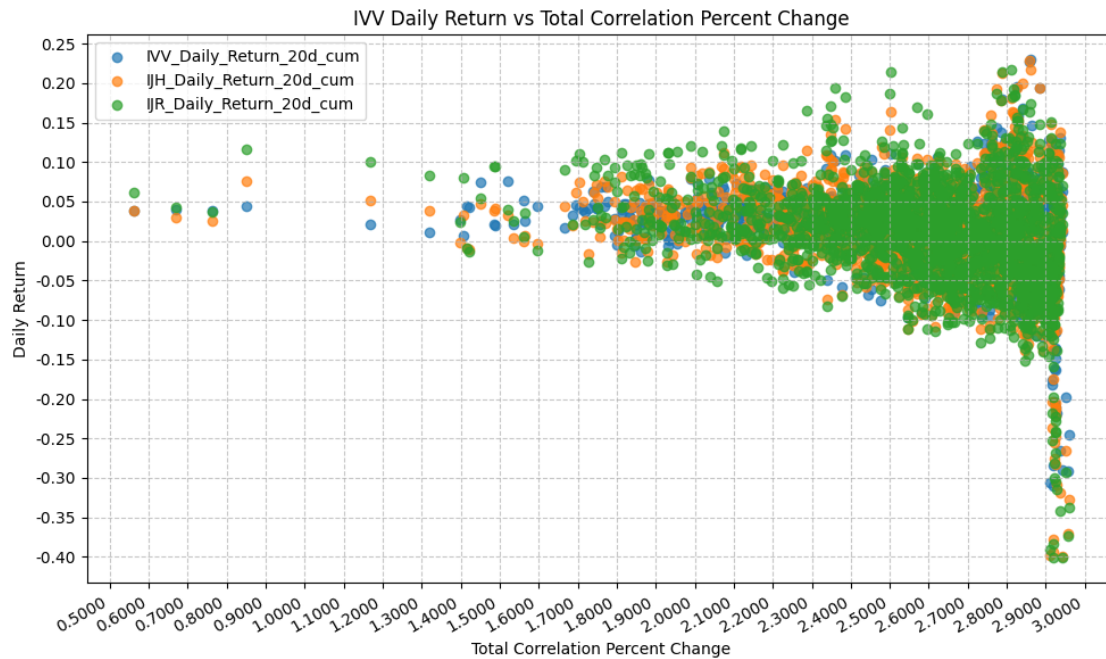
Date	IJR_Daily_Return_25d_cum
2006-08-21	NaN
2006-08-22	NaN
2006-08-23	NaN
2006-08-24	NaN
2006-08-25	NaN
...	...
2026-04-24	0.1102
2026-04-27	0.1341
2026-04-28	0.1050
2026-04-29	0.0886
2026-04-30	0.0961

[4954 rows x 28 columns]

```
[24]: plot_scatter(
    df=us_sp_etfs,
    x_plot_column="total_correlation",
    y_plot_columns=["IVV_Daily_Return_20d_cum", "IJH_Daily_Return_20d_cum",
↪ "IJR_Daily_Return_20d_cum"],
    title="IVV Daily Return vs Total Correlation Percent Change",
    x_label="Total Correlation Percent Change",
    x_format="Decimal",
    x_format_decimal_places=4,
    x_tick_spacing="Auto",
    x_tick_rotation=30,
    y_label="Daily Return",
    y_format="Decimal",
    y_format_decimal_places=2,
    y_tick_spacing="Auto",
    y_tick_rotation=0,
    grid=True,
    legend=True,
    export_plot=False,
    plot_file_name=None,
)
```

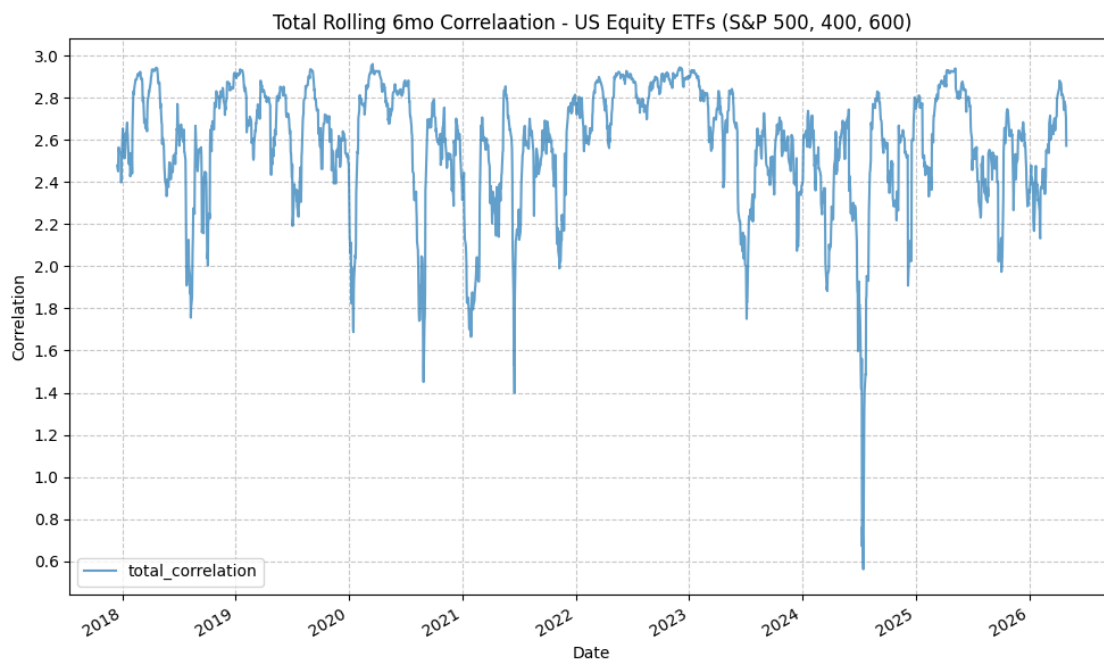
Raw x-tick spacing: 0.12050918913320856

Rounded x-tick spacing: 0.1



Now we'll plot the total correlation metric over time:

```
[25]: plot_time_series(
    df=us_sp_etfs.dropna(),
    plot_start_date=None,
    plot_end_date=None,
    plot_columns=["total_correlation"],
    title="Total Rolling 6mo Correlaation - US Equity ETFs (S&P 500, 400, 600)",
    x_label="Date",
    x_format="Year",
    x_tick_spacing=1,
    x_tick_rotation=30,
    y_label="Correlation",
    y_format="Decimal",
    y_format_decimal_places="Auto",
    y_tick_spacing="Auto",
    y_tick_rotation=0,
    grid=True,
    legend=True,
    export_plot=False,
    plot_file_name=None,
)
```



Now we'll look at a few different moving averages of the total correlation metric, the difference between the total correlation and the moving averages, and the percentage difference.

```
[26]: fund_data_daily_returns_all
```

```
[26]:      IVV_Daily_Return  IJH_Daily_Return  IJR_Daily_Return  \
```

Date	IVV_Daily_Return	IJH_Daily_Return	IJR_Daily_Return
1999-03-10	NaN	NaN	NaN
1999-03-11	NaN	NaN	NaN
1999-03-12	NaN	NaN	NaN
1999-03-15	NaN	NaN	NaN
1999-03-16	NaN	NaN	NaN
...	...	...	...
2026-04-27	0.0017	0.0007	0.0019
2026-04-28	-0.0049	-0.0100	-0.0056
2026-04-29	-0.0001	-0.0075	-0.0078
2026-04-30	0.0100	0.0168	0.0176
2026-05-01	NaN	NaN	NaN

Date	QQQ_Daily_Return	IWB_Daily_Return	IWM_Daily_Return
1999-03-10	NaN	NaN	NaN
1999-03-11	0.0049	NaN	NaN
1999-03-12	-0.0244	NaN	NaN
1999-03-15	0.0287	NaN	NaN
1999-03-16	0.0085	NaN	NaN

...	...	...	...
2026-04-27	0.0005	0.0011	0.0018
2026-04-28	-0.0101	-0.0053	-0.0117
2026-04-29	0.0061	-0.0005	-0.0067
2026-04-30	0.0093	0.0102	0.0216
2026-05-01	NaN	NaN	NaN

	IWD_Daily_Return	IWF_Daily_Return	EFA_Daily_Return	\
Date				
1999-03-10	NaN	NaN	NaN	
1999-03-11	NaN	NaN	NaN	
1999-03-12	NaN	NaN	NaN	
1999-03-15	NaN	NaN	NaN	
1999-03-16	NaN	NaN	NaN	

...	...	...	...
2026-04-27	0.0001	0.0017	-0.0038
2026-04-28	-0.0026	-0.0076	-0.0041
2026-04-29	-0.0000	-0.0023	-0.0102
2026-04-30	0.0184	0.0040	0.0239
2026-05-01	NaN	NaN	NaN

	EEM_Daily_Return	IEV_Daily_Return	SHY_Daily_Return	\
Date				
1999-03-10	NaN	NaN	NaN	
1999-03-11	NaN	NaN	NaN	
1999-03-12	NaN	NaN	NaN	
1999-03-15	NaN	NaN	NaN	
1999-03-16	NaN	NaN	NaN	

...	...	...	...
2026-04-27	-0.0016	-0.0052	-0.0002
2026-04-28	-0.0102	-0.0052	-0.0006
2026-04-29	-0.0048	-0.0116	-0.0013
2026-04-30	0.0207	0.0239	0.0011
2026-05-01	NaN	NaN	NaN

	IEF_Daily_Return	TLT_Daily_Return	AGG_Daily_Return	\
Date				
1999-03-10	NaN	NaN	NaN	
1999-03-11	NaN	NaN	NaN	
1999-03-12	NaN	NaN	NaN	
1999-03-15	NaN	NaN	NaN	
1999-03-16	NaN	NaN	NaN	

...	...	...	...
2026-04-27	-0.0023	-0.0050	-0.0015
2026-04-28	-0.0009	0.0010	-0.0004
2026-04-29	-0.0047	-0.0078	-0.0044
2026-04-30	0.0019	-0.0009	0.0014

2026-05-01	NaN	NaN	NaN
	GLD_Daily_Return	GSG_Daily_Return	IYR_Daily_Return \
Date			
1999-03-10	NaN	NaN	NaN
1999-03-11	NaN	NaN	NaN
1999-03-12	NaN	NaN	NaN
1999-03-15	NaN	NaN	NaN
1999-03-16	NaN	NaN	NaN
...	...	...	...
2026-04-27	-0.0078	0.0092	-0.0057
2026-04-28	-0.0186	0.0140	0.0090
2026-04-29	-0.0107	0.0366	-0.0083
2026-04-30	0.0150	-0.0049	0.0171
2026-05-01	NaN	NaN	NaN

	BTC-USD_Daily_Return	ETH-USD_Daily_Return
Date		
1999-03-10	NaN	NaN
1999-03-11	NaN	NaN
1999-03-12	NaN	NaN
1999-03-15	NaN	NaN
1999-03-16	NaN	NaN
...	...	...
2026-04-27	-0.0164	-0.0281
2026-04-28	-0.0131	-0.0059
2026-04-29	-0.0075	-0.0157
2026-04-30	0.0070	0.0013
2026-05-01	0.0246	0.0172

[8151 rows x 20 columns]

```
[27]: ma_windows = [5, 10, 15, 20, 25]

for window in ma_windows:
    us_sp_etfs[f"total_correlation_{window}d_ma"] =
    ↪us_sp_etfs["total_correlation"].rolling(window=window).mean()
    us_sp_etfs[f"total_correlation_diff_{window}d"] =
    ↪us_sp_etfs["total_correlation"] -
    ↪us_sp_etfs[f"total_correlation_{window}d_ma"]

display(us_sp_etfs)
```

	IVV_IJH_1mo	IVV_IJR_1mo	IJH_IJR_1mo	total_correlation \
Date				
2006-08-21	0.9585	0.9301	0.9665	2.8552
2006-08-22	0.9403	0.9066	0.9564	2.8034
2006-08-23	0.9462	0.9109	0.9574	2.8144

2006-08-24	0.9442	0.9206	0.9623	2.8271
2006-08-25	0.9515	0.9257	0.9578	2.8350
...	...	...	...	...
2026-04-24	0.8906	0.9159	0.9755	2.7820
2026-04-27	0.8768	0.9120	0.9730	2.7618
2026-04-28	0.8601	0.8967	0.9685	2.7253
2026-04-29	0.8529	0.8854	0.9677	2.7060
2026-04-30	0.7845	0.8284	0.9588	2.5717

Date	total_correlation_pct_change	IVV_Daily_Return	IJH_Daily_Return	\
2006-08-21	NaN	NaN	NaN	
2006-08-22	-0.0181	NaN	NaN	
2006-08-23	0.0040	NaN	NaN	
2006-08-24	0.0045	NaN	NaN	
2006-08-25	0.0028	NaN	NaN	
...	...	...	...	
2026-04-24	-0.0001	0.0078	0.0021	
2026-04-27	-0.0073	0.0017	0.0007	
2026-04-28	-0.0132	-0.0049	-0.0100	
2026-04-29	-0.0071	-0.0001	-0.0075	
2026-04-30	-0.0496	0.0100	0.0168	

Date	IJR_Daily_Return	total_correlation_pct_change_5d_cum	\
2006-08-21	NaN	NaN	
2006-08-22	NaN	NaN	
2006-08-23	NaN	NaN	
2006-08-24	NaN	NaN	
2006-08-25	NaN	NaN	
...	...	...	
2026-04-24	0.0054	-0.0104	
2026-04-27	0.0019	-0.0156	
2026-04-28	-0.0056	-0.0223	
2026-04-29	-0.0078	-0.0132	
2026-04-30	0.0176	-0.0757	

Date	IVV_Daily_Return_5d_cum	...	total_correlation_5d_ma	\
2006-08-21	NaN	...	NaN	
2006-08-22	NaN	...	NaN	
2006-08-23	NaN	...	NaN	
2006-08-24	NaN	...	NaN	
2006-08-25	NaN	...	2.8270	
...	...	...	...	
2026-04-24	0.0055	...	2.7798	
2026-04-27	0.0090	...	2.7711	
2026-04-28	0.0109	...	2.7587	

2026-04-29	0.0005 ...	2.7514
2026-04-30	0.0145 ...	2.7093

Date	total_correlation_diff_5d	total_correlation_10d_ma \
2006-08-21	NaN	NaN
2006-08-22	NaN	NaN
2006-08-23	NaN	NaN
2006-08-24	NaN	NaN
2006-08-25	0.0080	NaN
...	...	...
2026-04-24	0.0021	2.8053
2026-04-27	-0.0093	2.7948
2026-04-28	-0.0334	2.7828
2026-04-29	-0.0455	2.7723
2026-04-30	-0.1376	2.7475

Date	total_correlation_diff_10d	total_correlation_15d_ma \
2006-08-21	NaN	NaN
2006-08-22	NaN	NaN
2006-08-23	NaN	NaN
2006-08-24	NaN	NaN
2006-08-25	NaN	NaN
...	...	...
2026-04-24	-0.0234	2.8245
2026-04-27	-0.0331	2.8195
2026-04-28	-0.0575	2.8115
2026-04-29	-0.0663	2.7999
2026-04-30	-0.1758	2.7795

Date	total_correlation_diff_15d	total_correlation_20d_ma \
2006-08-21	NaN	NaN
2006-08-22	NaN	NaN
2006-08-23	NaN	NaN
2006-08-24	NaN	NaN
2006-08-25	NaN	NaN
...	...	...
2026-04-24	-0.0426	2.8094
2026-04-27	-0.0577	2.8125
2026-04-28	-0.0862	2.8131
2026-04-29	-0.0939	2.8087
2026-04-30	-0.2077	2.7973

Date	total_correlation_diff_20d	total_correlation_25d_ma \
2006-08-21	NaN	NaN

2006-08-22	NaN	NaN
2006-08-23	NaN	NaN
2006-08-24	NaN	NaN
2006-08-25	NaN	NaN
...	...	...
2026-04-24	-0.0275	2.7801
2026-04-27	-0.0507	2.7837
2026-04-28	-0.0878	2.7847
2026-04-29	-0.1027	2.7871
2026-04-30	-0.2255	2.7842

total_correlation_diff_25d	
Date	
2006-08-21	NaN
2006-08-22	NaN
2006-08-23	NaN
2006-08-24	NaN
2006-08-25	NaN
...	...
2026-04-24	0.0019
2026-04-27	-0.0220
2026-04-28	-0.0594
2026-04-29	-0.0811
2026-04-30	-0.2124

[4954 rows x 38 columns]

```
[28]: us_sp_etfs[(us_sp_etfs.index >= "2020-01-01") & (us_sp_etfs.index <=
↪ "2021-01-01")]
```

	IVV_IJH_1mo	IVV_IJR_1mo	IJH_IJR_1mo	total_correlation	\
Date					
2020-01-02	0.8117	0.5723	0.8445	2.2285	
2020-01-03	0.8043	0.5143	0.8195	2.1381	
2020-01-06	0.7826	0.4706	0.8061	2.0593	
2020-01-07	0.7916	0.5078	0.8126	2.1120	
2020-01-08	0.7544	0.4078	0.7530	1.9152	
...	...	...	...	...	
2020-12-24	0.8284	0.7455	0.9679	2.5418	
2020-12-28	0.7470	0.7170	0.9549	2.4188	
2020-12-29	0.7477	0.7148	0.9585	2.4210	
2020-12-30	0.7293	0.6992	0.9490	2.3774	
2020-12-31	0.6849	0.6787	0.9511	2.3146	

	total_correlation_pct_change	IVV_Daily_Return	IJH_Daily_Return	\
Date				
2020-01-02	-0.1028	0.0095	0.0015	

2020-01-03	-0.0405	-0.0077	-0.0050
2020-01-06	-0.0369	0.0040	-0.0003
2020-01-07	0.0256	-0.0027	-0.0026
2020-01-08	-0.0932	0.0051	0.0016
...	...	...	...
2020-12-24	-0.0276	0.0039	0.0016
2020-12-28	-0.0484	0.0087	-0.0028
2020-12-29	0.0009	-0.0019	-0.0103
2020-12-30	-0.0180	0.0013	0.0087
2020-12-31	-0.0264	0.0056	0.0008

Date	IJR_Daily_Return	total_correlation_pct_change_5d_cum \
2020-01-02	0.0007	-0.1177
2020-01-03	-0.0006	-0.1507
2020-01-06	-0.0010	-0.1875
2020-01-07	-0.0057	-0.1514
2020-01-08	0.0022	-0.2289
...	...	...
2020-12-24	0.0009	-0.0390
2020-12-28	0.0038	-0.0745
2020-12-29	-0.0171	-0.0775
2020-12-30	0.0101	-0.0951
2020-12-31	0.0015	-0.1146

Date	IVV_Daily_Return_5d_cum	...	total_correlation_5d_ma \
2020-01-02	0.0114	...	2.4506
2020-01-03	-0.0016	...	2.3747
2020-01-06	0.0026	...	2.2797
2020-01-07	0.0052	...	2.2043
2020-01-08	0.0081	...	2.0906
...	...	...	...
2020-12-24	-0.0044	...	2.6042
2020-12-28	0.0082	...	2.5653
2020-12-29	0.0096	...	2.5246
2020-12-30	0.0129	...	2.4746
2020-12-31	0.0176	...	2.4147

Date	total_correlation_diff_5d	total_correlation_10d_ma \
2020-01-02	-0.2221	2.5055
2020-01-03	-0.2366	2.4589
2020-01-06	-0.2204	2.4043
2020-01-07	-0.0923	2.3618
2020-01-08	-0.1754	2.3003
...	...	...

2020-12-24	-0.0624	2.6308
2020-12-28	-0.1465	2.6026
2020-12-29	-0.1036	2.5777
2020-12-30	-0.0972	2.5487
2020-12-31	-0.1001	2.5198

Date	total_correlation_diff_10d	total_correlation_15d_ma \
2020-01-02	-0.2770	2.5434
2020-01-03	-0.3207	2.5130
2020-01-06	-0.3451	2.4742
2020-01-07	-0.2498	2.4396
2020-01-08	-0.3851	2.3920
...	...	...
2020-12-24	-0.0889	2.6132
2020-12-28	-0.1838	2.6015
2020-12-29	-0.1568	2.5916
2020-12-30	-0.1713	2.5785
2020-12-31	-0.2052	2.5643

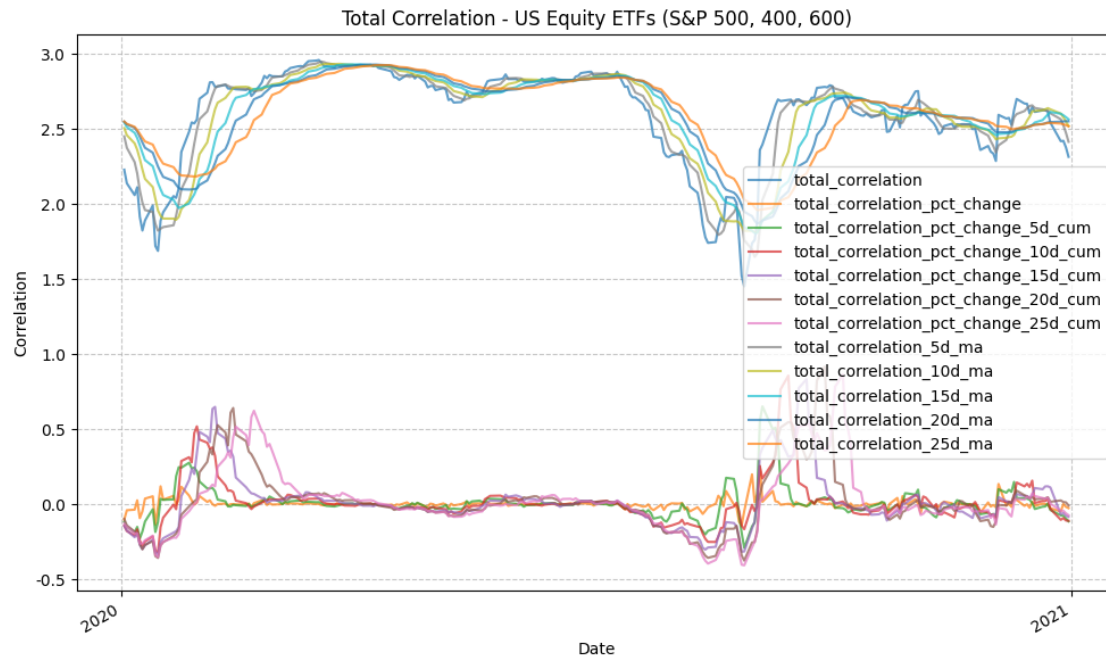
Date	total_correlation_diff_15d	total_correlation_20d_ma \
2020-01-02	-0.3150	2.5487
2020-01-03	-0.3748	2.5311
2020-01-06	-0.4150	2.5081
2020-01-07	-0.3277	2.4837
2020-01-08	-0.4769	2.4491
...	...	...
2020-12-24	-0.0714	2.5459
2020-12-28	-0.1827	2.5488
2020-12-29	-0.1707	2.5505
2020-12-30	-0.2011	2.5519
2020-12-31	-0.2497	2.5508

Date	total_correlation_diff_20d	total_correlation_25d_ma \
2020-01-02	-0.3202	2.5495
2020-01-03	-0.3930	2.5343
2020-01-06	-0.4489	2.5143
2020-01-07	-0.3717	2.4940
2020-01-08	-0.5339	2.4668
...	...	...
2020-12-24	-0.0040	2.5405
2020-12-28	-0.1300	2.5360
2020-12-29	-0.1295	2.5316
2020-12-30	-0.1745	2.5254
2020-12-31	-0.2362	2.5180

	total_correlation_diff_25d
Date	
2020-01-02	-0.3210
2020-01-03	-0.3961
2020-01-06	-0.4550
2020-01-07	-0.3821
2020-01-08	-0.5516
...	...
2020-12-24	0.0014
2020-12-28	-0.1172
2020-12-29	-0.1106
2020-12-30	-0.1480
2020-12-31	-0.2034

[253 rows x 38 columns]

```
[29]: plot_time_series(
        df=us_sp_etfs[(us_sp_etfs.index >= "2020-01-01") & (us_sp_etfs.index <=
        ↪ "2021-01-01")],
        plot_start_date=None,
        plot_end_date=None,
        plot_columns=[col for col in us_sp_etfs.columns if "total_correlation" in
        ↪ col and "diff" not in col],
        title="Total Correlation - US Equity ETFs (S&P 500, 400, 600)",
        x_label="Date",
        x_format="Year",
        x_tick_spacing=1,
        x_tick_rotation=30,
        y_label="Correlation",
        y_format="Decimal",
        y_format_decimal_places="Auto",
        y_tick_spacing="Auto",
        y_tick_rotation=0,
        grid=True,
        legend=True,
        export_plot=False,
        plot_file_name=None,
    )
```



```
[30]: fund_data_daily_returns_all
```

```
[30]:
```

	IVV_Daily_Return	IJH_Daily_Return	IJR_Daily_Return	\
Date				
1999-03-10	NaN	NaN	NaN	
1999-03-11	NaN	NaN	NaN	
1999-03-12	NaN	NaN	NaN	
1999-03-15	NaN	NaN	NaN	
1999-03-16	NaN	NaN	NaN	
...	...	...	...	
2026-04-27	0.0017	0.0007	0.0019	
2026-04-28	-0.0049	-0.0100	-0.0056	
2026-04-29	-0.0001	-0.0075	-0.0078	
2026-04-30	0.0100	0.0168	0.0176	
2026-05-01	NaN	NaN	NaN	

	QQQ_Daily_Return	IWB_Daily_Return	IWM_Daily_Return	\
Date				
1999-03-10	NaN	NaN	NaN	
1999-03-11	0.0049	NaN	NaN	
1999-03-12	-0.0244	NaN	NaN	
1999-03-15	0.0287	NaN	NaN	
1999-03-16	0.0085	NaN	NaN	
...	...	...	...	
2026-04-27	0.0005	0.0011	0.0018	

2026-04-28	-0.0101	-0.0053	-0.0117
2026-04-29	0.0061	-0.0005	-0.0067
2026-04-30	0.0093	0.0102	0.0216
2026-05-01	NaN	NaN	NaN

	IWD_Daily_Return	IWF_Daily_Return	EFA_Daily_Return	\
Date				
1999-03-10	NaN	NaN	NaN	
1999-03-11	NaN	NaN	NaN	
1999-03-12	NaN	NaN	NaN	
1999-03-15	NaN	NaN	NaN	
1999-03-16	NaN	NaN	NaN	
...	...	...	...	
2026-04-27	0.0001	0.0017	-0.0038	
2026-04-28	-0.0026	-0.0076	-0.0041	
2026-04-29	-0.0000	-0.0023	-0.0102	
2026-04-30	0.0184	0.0040	0.0239	
2026-05-01	NaN	NaN	NaN	

	EEM_Daily_Return	IEV_Daily_Return	SHY_Daily_Return	\
Date				
1999-03-10	NaN	NaN	NaN	
1999-03-11	NaN	NaN	NaN	
1999-03-12	NaN	NaN	NaN	
1999-03-15	NaN	NaN	NaN	
1999-03-16	NaN	NaN	NaN	
...	...	...	...	
2026-04-27	-0.0016	-0.0052	-0.0002	
2026-04-28	-0.0102	-0.0052	-0.0006	
2026-04-29	-0.0048	-0.0116	-0.0013	
2026-04-30	0.0207	0.0239	0.0011	
2026-05-01	NaN	NaN	NaN	

	IEF_Daily_Return	TLT_Daily_Return	AGG_Daily_Return	\
Date				
1999-03-10	NaN	NaN	NaN	
1999-03-11	NaN	NaN	NaN	
1999-03-12	NaN	NaN	NaN	
1999-03-15	NaN	NaN	NaN	
1999-03-16	NaN	NaN	NaN	
...	...	...	...	
2026-04-27	-0.0023	-0.0050	-0.0015	
2026-04-28	-0.0009	0.0010	-0.0004	
2026-04-29	-0.0047	-0.0078	-0.0044	
2026-04-30	0.0019	-0.0009	0.0014	
2026-05-01	NaN	NaN	NaN	

	GLD_Daily_Return	GSG_Daily_Return	IYR_Daily_Return	\
Date				
1999-03-10	NaN	NaN	NaN	
1999-03-11	NaN	NaN	NaN	
1999-03-12	NaN	NaN	NaN	
1999-03-15	NaN	NaN	NaN	
1999-03-16	NaN	NaN	NaN	
...	...	...	...	
2026-04-27	-0.0078	0.0092	-0.0057	
2026-04-28	-0.0186	0.0140	0.0090	
2026-04-29	-0.0107	0.0366	-0.0083	
2026-04-30	0.0150	-0.0049	0.0171	
2026-05-01	NaN	NaN	NaN	

	BTC-USD_Daily_Return	ETH-USD_Daily_Return
Date		
1999-03-10	NaN	NaN
1999-03-11	NaN	NaN
1999-03-12	NaN	NaN
1999-03-15	NaN	NaN
1999-03-16	NaN	NaN
...	...	...
2026-04-27	-0.0164	-0.0281
2026-04-28	-0.0131	-0.0059
2026-04-29	-0.0075	-0.0157
2026-04-30	0.0070	0.0013
2026-05-01	0.0246	0.0172

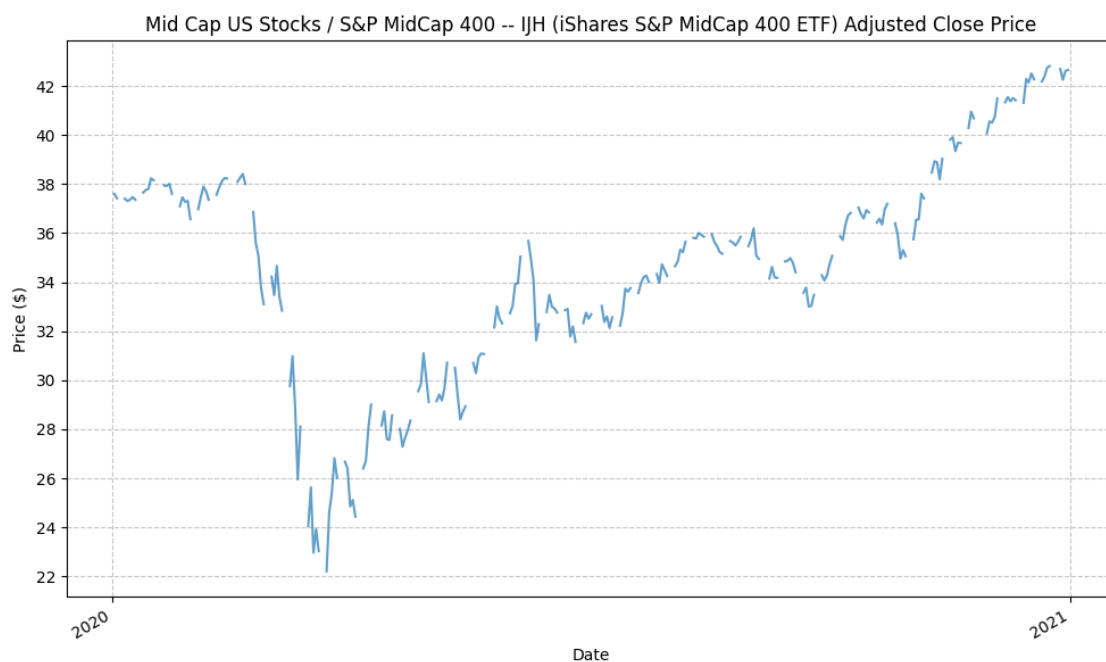
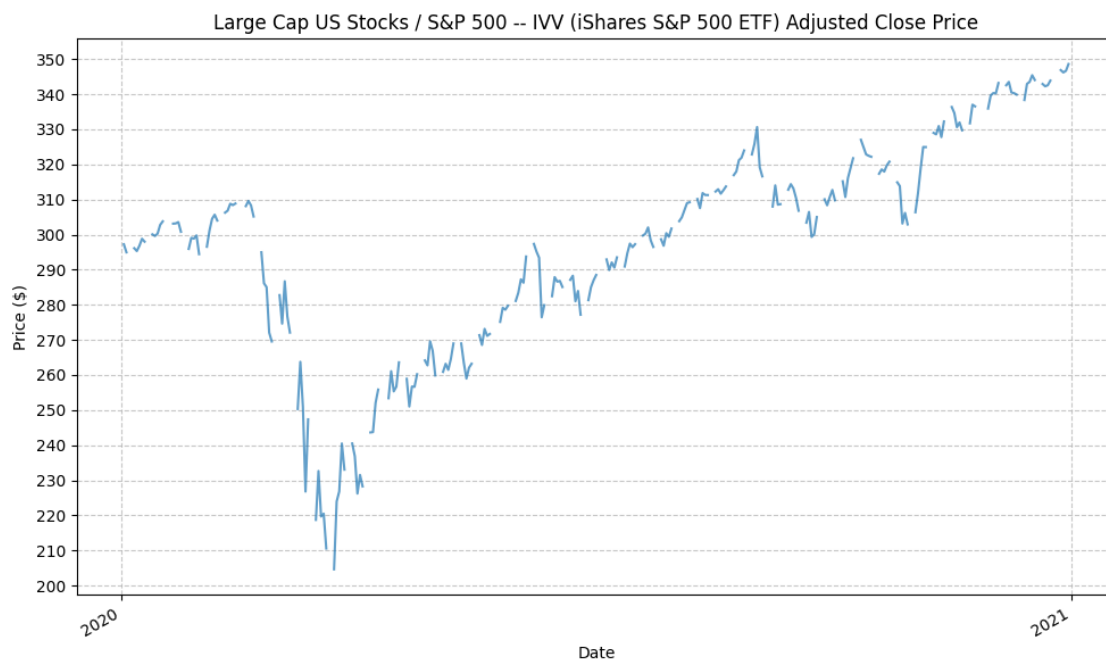
[8151 rows x 20 columns]

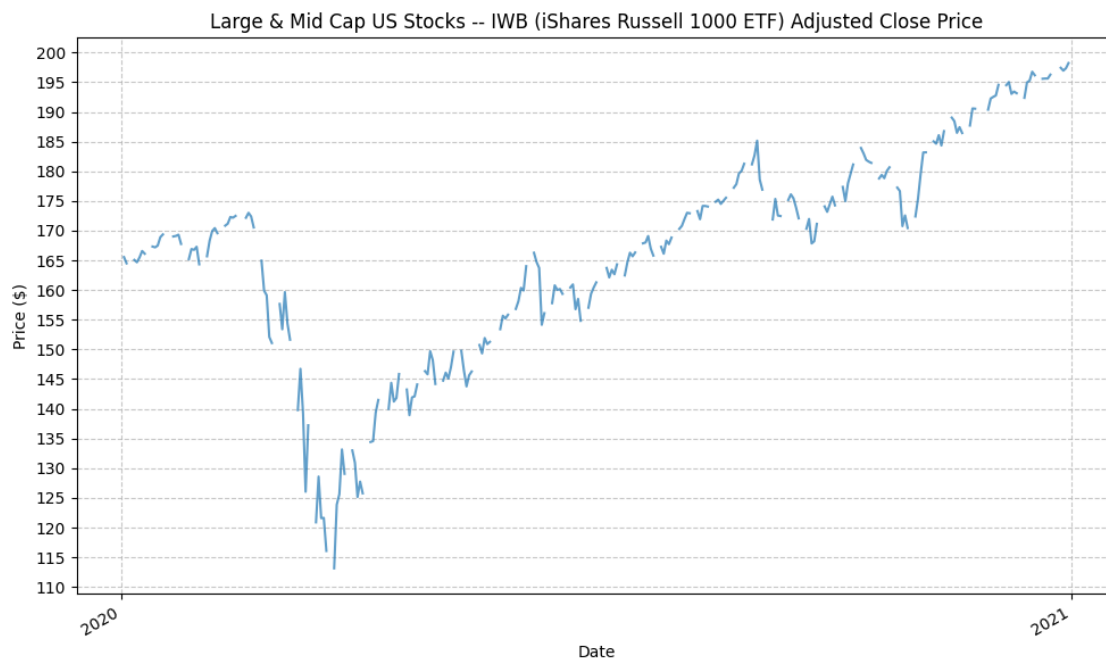
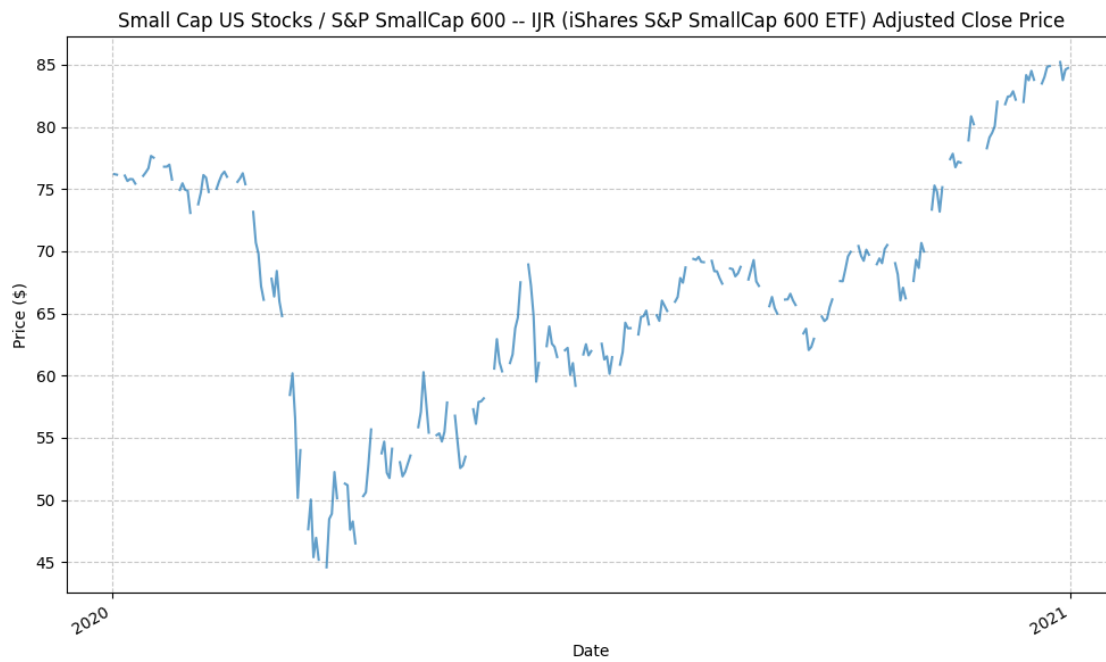
```
[31]: for ticker in us_equity_tickers:
        if ticker not in ["QQQ", "IWD", "IWF"]:
            plot_time_series(
                df=fund_data[(fund_data.index >= "2020-01-01") & (fund_data.index_
↵ <= "2021-01-01")],
                plot_start_date=None,
                plot_end_date=None,
                plot_columns=[f"{ticker}_Adj_Close"],
                title=f"{tickers_dict[ticker]} Adjusted Close Price",
                x_label="Date",
                x_format="Year",
                x_tick_spacing=1,
                x_tick_rotation=30,
                y_label="Price ($)",
                y_format="Decimal",
                y_format_decimal_places="Auto",
                y_tick_spacing="Auto",
```

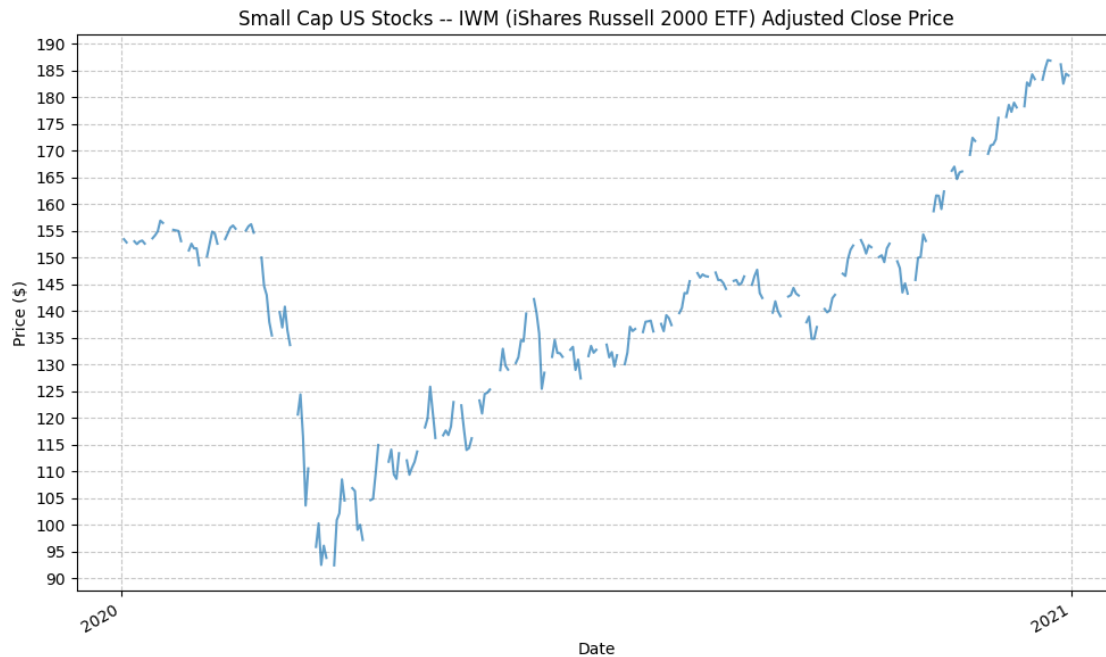
```

y_tick_rotation=0,
grid=True,
legend=False,
export_plot=False,
plot_file_name=None,
)

```







Now that we have the plots, we can proceed with an analysis of the rolling correlations and attempt to tie the changes in the rolling correlations to specific market events (e.g. the 2008 financial crisis, the COVID-19 pandemic, etc.) or broader market trends. Essentially, we want to find out if there are any interesting market trends or regimes that are precipitated by changes in the rolling correlations between asset classes. It's a common belief that correlations between assets classes increase (go towards 1) during times of market stress, so we will be looking to see if this is the case and if so does this change in correlation in any way predict market returns.

0.7 Future Investigation

None for now.

0.8 References

None for now.